FINAL REPORT - SOUTHERN HEMISPHERE SPACE STUDIES PROGRAM 2023

# JOURNEY TO THE MOON

## TECHNOLOGICAL AND LEGAL PERSPECTIVES FOR SUSTAINABLE HUMAN PRESENCE



INTERNATIONAL SPACE UNIVERSITY
ISU
University of South Australia

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International Space University
Southern Hemisphere Space Studies Program 2023
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We sincerely appreciate this opportunity and have given our best to rise to the challenge.
We hope you enjoy reading our vision.

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# Abstract

Returning to the Moon and establishing a permanent human presence is the next step in space exploration. Achieving this presence on the lunar surface will require a collective effort from all humankind, due to the challenges related to its exploration that will appear both in terms of technologies and international interactions between various countries and players involved.

During the Southern Hemisphere Space Studies Program (SHSSP23), this team has identified the most pressing of these challenges, looking for practical solutions that would ensure a sustainable presence of the human species on the Moon. The concept of sustainability, defined as the ability to maintain a continuous long-term human program on the Moon, and how to ensure it, is the central focus around which this work is developed. Among all the factors required to achieve sustainability, we focus on the requirements which relate most strongly to the physical sustainability of a lunar settlement: power, in-situ resource utilization (ISRU), and human habitation. Since the achievability of a sustainable human presence is directly correlated with the technology and research available, this team project addresses the challenge of sustainability by considering the following roadmap: near-term (until 2030), mid-term (until 2040) and long-term (until 2050). Thus, for each of these three requirements, a detailed presentation on the current status of the technologies developed is provided, presenting the challenges in sustained human settlement on the Moon, and giving recommendations for their future use in all the various phases, for a beneficial human presence on the lunar surface. A roadmap and a series of tables summarizing the contents and the information defined are also present in the report. The constructive discussions conducted by the team have highlighted the gaps in the current international legal framework of proper means to tackle the most pressing contemporary issues surrounding expected activity in cislunar space and on the Moon. The international legal instruments that regulate outer space have been revised proposing a new version of the international treaties, based on the Artemis Accords and the Antarctic Treaty, and picturing a two-tiered system with which to handle the international interaction among different parties, both governments and private entities, in lunar activities.

At the end of this report, a series of comprehensive and conclusive recommendations, both from a technological and legal point of view, have been addressed. We hope that the same doubts and questions we have been asking ourselves will arise in the future in some other institution, and they may find in this work a solid base for their considerations for peaceful and sustainable lunar exploration.

# Faculty Preface

This report examines the future of activities on the Moon. Fifty years after the Apollo missions, preparations are being made for humans to return to the Moon. The international Artemis program, led by NASA, has completed Artemis I, the first uncrewed test mission, and is planning on landing astronauts near the lunar south pole in a few years. In addition, a partnership between China and Russia is planning the International Lunar Research Station, also intended to be located near the lunar south pole. Many independent national and international government lunar programs are occurring in parallel. And commercial lunar companies are about to initiate operations on the Moon. A great deal of public and private investment is occurring, but without much long-term coordination or planning.

How should these lunar activities be coordinated? How can lunar activities be sustained for long-term human presence while protecting opportunities for the future? These were the challenges given to the authors of this report.

The Southern Hemisphere Space Studies Program (SHSSP) of the International Space University (ISU) is an intensive five-week program covering all aspects of space. The SHSSP23 program was hosted at the Mawson Lakes campus of the University of South Australia in Adelaide, Australia.

The team project is an important culmination of the program requiring participants to integrate their interdisciplinary lessons and apply them to a specific topic. This year, 19 participants from ten countries contributed to this report. The participants had expertise ranging from physics, engineering, remote sensing, information technology, medicine, life sciences, business, management, to policy and law. This diverse team examined how to guide the future of activities on the Moon. They identified technical roadmaps for critical aspects of long-term human presence on the Moon, and proposed an international legal framework for coordinating activities from multiple nations and stakeholders.

We hope the experience of the ISU SHSSP23 participants in this international study, and the results captured in this report, will help guide us toward a future of peaceful lunar activities.

Eric Dahlstrom, Team Project Chair

# Participant Preface

Humanity plans to return to the Moon in the next decade for the beginning of a permanent settlement. Many nations, their agencies and private companies are targeting the same areas for landing which present a lot of challenges, but also potential for collaboration.

We were invited to address these challenges and decided to take a step back and evaluate the global scenario. Instead of following a single lunar program, the group found that it would be interesting to have a historical analysis of past programs and a preview of future Moon activities from all stakeholders involved. Our vision is to contribute to the long-term sustainable human presence on the Moon.

Due to the limited amount of time, we decided to focus the scope of research on the resources necessary to get Humanity on the Moon and proposed a legal framework to achieve it. The team is aware that this report provides an initial document that can and should be expanded in future research.

We hope to leave you inspired and that the reader now thinks of the Earth-Moon equation with a global perspective. This group chose this topic because we are all interested in expanding the humanity's footprint in the solar system; Journey to the Moon that explores the future with curiosity.

Team Journey to the Moon,

February 17, 2023

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# List of Acronyms

| Acronym | Meaning |
|---|---|
| AB | Appellate Body |
| ABC | Artemis Base Camp |
| AI | Artificial Intelligence |
| AM | Additive Manufacturing |
| ATS | Antarctic Treaty System |
| BRIC | Brazil Russia India China |
| CLEP | Chinese Lunar Exploration Program |
| CNSA | China National Space Administration |
| CNSA | Chinese National Space Agency |
| COLA | Committee Overseeing Lunar Activities |
| COPUOS | Committee on the Peaceful Uses of Outer Space |
| CPV | Concentrator Photovoltaics |
| CSA | Canadian Space Agency |
| EM | Electromagnetic |
| ESA | European Space Agency |
| EU | European Union |
| EUSPA | EU Agency for the Space Programme |
| FFC | Fray-Farthing-Chen |
| FPS | Fission Power System |
| FSH | Foundational Surface Habitat |
| GRAIL | Gravity Recovery and Interior Laboratory |
| H/H2 | Hydrogen/Molecular hydrogen |
| HALO | Habitation and Logistics Outpost |
| HCl | Hydrogen chloride |
| HLS | Human Landing System |
| HMP | Habitable Mobility Platform |
| ICJ | International Court of Justice |
| ICoC | International Code of Conduct |
| IGA | ISS Intergovernmental Agreement |
| I-HAB | International Habitation module |
| ILO | International Labour Organization |
| ILRS | International Lunar Research Station |
| IPU | Inter-Parliamentary Union |
| ISM | In Space Manufacturing |
| ISRO | Indian Space Research Organisation |
| ISRU | In-Situ Resource Utilization |
| ISS | International Space Station |
| ITA | International Telecommunications Agreement |
| JAXA | Japan Aerospace Exploration Agency |
| LAT | Lunar Activities Treaty |
| LCROSS | Lunar Crater Observation and Sensing Satellite |
| LEM | Lunar Excursion Module |
| LEO | Low Earth Orbit |
| Li | Lithium |

| | |
|---|---|
| LPT | Laser Power Transmission |
| LRO | Lunar Reconnaissance Orbiter |
| LTV | Lunar Terrain Vehicle |
| LUPEX | Lunar Polar Exploration Mission |
| MG | Micro Grid |
| MMRTG | Multi-Mission Radioisotope Thermoelectric Generator |
| MoU | Memorandum of Understanding |
| MPT | Microwave Power Transmission |
| MVA | Moon Village Association |
| Na | Sodium |
| NASA | National Aeronautics and Space Administration |
| Ni | Nickel |
| NRHO | Near-Rectilinear Halo Orbit |
| OST | Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, 1967 |
| PMAD | Power Management and Distribution |
| PPE | Power and Propulsion Element |
| PPP | Public–Private Partnership |
| PSR | Permanently Shadowed Region |
| Pu | Plutonium |
| PVA | Photovoltaic Arrays |
| RHU | Radioisotope Heater Units |
| RPS | Radioisotope Power System |
| RTG | Radioisotope Thermoelectric Generators |
| SELENE | Selenological and Engineering Explorer |
| SHSSP23 | Southern Hemisphere Space Studies Program 2023 |
| SLIM | Smart Lander for Investigating Moon |
| SLS | Space Launch System |
| SMART-1 | Small Missions for Advanced Research in Technology-1 |
| STEM | Science, Technology, Engineering and Mathematics |
| TRL | Technological Readiness Level |
| U | Uranium |
| UN | United Nations |
| UNGA | United Nations General Assembly |
| UNOOSA | United Nations Office for Outer Space Affairs |
| USA | United States of America |
| WTO | World Trade Organization |

# 1 General Introduction

Fifty years after the last human exploration of the lunar surface, the success of the National Aeronautics and Space Administration's (NASA) Artemis I mission in December 2022 showcased determination, perseverance and vision, in returning humans to the Moon by the end of this decade to establish a permanent outpost in deep space.

The Southern Hemisphere Space Studies Program 2023 cohort seeks to contribute to the global discussion about how this return to the Moon can be best developed to achieve a peaceful and long-term presence. This report is the culmination of an intensive, interdisciplinary research project on the following theme: "Identification of issues and problems in lunar exploration, in order to define how to achieve a long-term human presence on the Moon, with a focus on technological challenges of resourcing and the legal aspects required for ensuring a peaceful international cooperation".

This report explores different resolutions for the complex intricacies of future cislunar activity, presenting the technological issues that have to be addressed in the next 30 years up to 2050, and illustrating the major concerns that derive from them. The focus has been on the handling of certain requirements which relate most strongly to the physical sustainability of a lunar settlement: power, in-situ resource utilization (ISRU), and human habitation. In terms of timeline, this team project addresses the challenge of sustainability by considering the following roadmap: near-term (until 2030), mid-term (until 2040) and long-term (until 2050). For each of the three requirements identified, a detailed analysis on the current status of the technologies developed is provided, presenting the challenges in sustained human settlement on the Moon, and giving recommendations for their future use in each of the defined phases, for a beneficial human presence on the lunar surface.

Today's challenges of long-term sustainable life in the harsh environment of cislunar space and on the surface of the Moon are driven by the combined efforts of private and public space organizations. This illustrates that international cooperation and collective effort are required to pursue the lunar exploration with a peaceful and cooperative spirit. Accordingly, this report also highlights the necessity for a global legal framework and presents a possible model. In particular, the focus of this report is on how humans should aim for a sustainable exploitation of cislunar space and of lunar resources, where for the purpose of this report sustainability is defined as the ability to maintain a continuous long-term human presence on the Moon. Thus, the legal framework created will ensure that the future of human lunar presence, and space resource utilization are pursued in a sustainable manner for all humankind.

The study is divided into three main sections, covering the three topics of historical legacy, resources utilization for long-term settlement on the Moon, and legal framework.

**Section 3**. This section is a historical overview of lunar exploration in the last 70 years and how it has evolved from the period of the Cold War, with only the United States of America (USA) and the Soviet Union involved, up to the present, with government agencies from many different countries as well as private actors like SpaceX or Blue Origin. After this, the status of the major government lunar exploration programs, such as Artemis and the International Lunar Research Station, and of the private commercial programs are illustrated, describing past and present efforts, and highlighting the challenges of competition demonstrated by various roadmaps.

**Section 4**. This section addresses the path to sustainable lunar settlements focusing on the factors required to achieve sustainability. Three requirements have been identified as those which relate most strongly to the physical sustainability of a lunar settlement: power, ISRU and human habitation. The challenge associated with each are addressed by considering the following roadmap: near-term (until 2030), mid-term (until 2040) and long-term (until 2050). For each requirement, a detailed description of the current status of the technologies developed is provided, presenting the challenges in sustained human settlement on the Moon. Recommendations for the future use of these technologies are also given.

**Section 5**. This section highlights and analyzes the current legal landscape that would apply to cislunar space. As the current space law is deemed to be insufficient, a legal probe is initiated to find a working model to govern such activities. The instruments that apply to outer space are used as a placeholder alongside other international treaties such as the Antarctic Treaty, the Intergovernmental Agreement of the International Space Station, and the UN Charter to draft a new model treaty. Further, an international organization along the lines of the United Nations, the International Labour Organization and the World Trade Organization, is also proposed to manage issues in cislunar space and to make sure that each stakeholder gets representation. This two-tier structure is suggested as a model for the regulation of future activities in cislunar space and on the Moon.

# 2 Scope of the Project

This report explores some of the optimal ways to navigate the complexities of future cislunar activity. In the current scenario, lunar exploration and human settlement on the Moon is a primary goal. To achieve this, international cooperation and collective effort to overcome technological challenges is critical. A universal legal framework is essential to allow human presence and utilization of resources on the Moon for all humankind.

The scope of this report is limited to providing recommendations for the development and maintenance of a lunar settlement that respects the principles of sustainability. These recommendations cover the three selected areas and are discussed in details throughout the report. Extensive research has been done to produce this report, so only highlights have been recorded. The project aims in short are as follows:

1. To analyze and communicate the varying rationales for returning to the Moon and developing a permanent presence.
2. To set out a roadmap consisting of a multiphasic approach to the development of a lunar settlement.
3. To develop a legal framework ensuring the respect of sustainability when embarking on future lunar missions.

To achieve these project aims, the following objectives were defined:

1. Review current lunar base plans and develop a consolidated, generic, five phase concept for lunar settlement evolution.
2. Evaluate said concept to identify technical and legal gaps that hinder sustainable development of the Moon.
3. Generate recommendations to ensure the sustainable development of the Moon.

# 3 Evolution of Lunar Exploration

## 3.1 Introduction

Lunar exploration has fascinated humanity for decades. Since the first Soviet exploration mission was launched in 1959, the Moon has been the subject of numerous investigations aimed at discovering new information about our natural satellite and its future utilization.

Current lunar exploration programs like Artemis and the International Lunar Research Station (ILRS) have evolved significantly from the early days of spaceflight. While early programs were predominantly driven by prestige and political ambitions, current programs have more practical and focused goals, such as scientific research, development of advanced technologies for space exploration, collection of valuable resources, and future use of the Moon as a base for further exploration in space.

In addition, there are commercial programs that aim to establish a continued presence on the Moon, providing services and infrastructure for future space activities, and the development of advanced technologies. These programs also aim to create economic and commercial opportunities related to lunar exploration.

This section provides a brief historical overview of lunar exploration, current government, and commercial ventures, and how they compare. Moreover, the differences between these programs in terms of goals, technologies, and future ambitions will be explored. Finally, the geopolitical implications of all the various planned lunar exploration missions are considered.

## 3.2 Historical Lunar Exploration

Human exploration of space has been gradually advancing since the first flights in 1959. Beginning with the Soviet Union and the USA, many countries have since contributed to its progress. Each major effort realized since that moment is highlighted in this section, with a particular focus on American, Soviet, Chinese, and Indian contributions.

During the decade between 1959 and 1969 both the Soviet Union and the USA sent several uncrewed probes for the purpose of studying the lunar surface and preparing for future missions to the Moon. The first success occurred in October 1959, when the Soviet Luna 3 spacecraft first photographed the far side of the Moon, shown in figure 1 (Kohonen, 2009). In the Cold War, the competition between the USA and the Soviet Union initiated various space programs in those years to photograph the lunar surface. With the American Ranger 7 probe, launched in July 1964, more than 4,000 photographs of the Moon were obtained (Brinkmann, 1966).



*Figure 1: The first photo of the Moon's far side returned by Luna 3. Credit NASA*

However, during all these early launches, due to a lack of technological development, probes were unable to land softly on the Moon, and crashed on the lunar surface after successfully photographing it. The first soft landing on the Moon was achieved by the Soviet Union's Luna 9 probe in 1966. In the same year, another important milestone was achieved by the Soviets with their Luna 10 craft managing to make an orbit around Earth's satellite (Efanov and Dolgopolov, 2017). Another first by the Soviet Union was the landing of Lunokhod 1, the first rover to operate on the lunar surface (Malenkov, 2016).

From the Western perspective, the American Surveyor space probes, launched between 1966 and 1968, were able to make controlled landings on the lunar surface. They were equipped with cameras to monitor the ground, and instruments that could analyze the rocks and lunar soil (Crawford and Joy, 2014). In 1966 and 1967 NASA launched their first orbiters, designed to orbit the Moon and map its surface in anticipation of future crewed missions (Lorell and Sjogren, 1968).

Between 1961 and 1972, NASA developed the Apollo program, which involved a crew of three astronauts aboard a space probe consisting of three modules: the command module, the service module, and the Lunar Module, originally designated the Lunar Excursion Module (LEM) - which served to land two astronauts on the Moon while the third remained in lunar orbit (Kelly, 2009). The Apollo program began with a major accident on January 27, 1967: a fire aboard Apollo 1 during an exercise, that killed three astronauts (DiLisi and McLean, 2019). Following this tragedy, NASA successfully launched several uncrewed missions, Apollos 4, 5, and 6 (HARDY, 1969). Aboard Apollo 7, launched in October 1968, a crew of astronauts remained in Earth orbit for more than eleven days (Berry, 1970). On December 21, 1968, the Apollo 8 mission was launched, performing the first crewed lunar orbit (Kluger, 2017). Apollo 9, in March 1969 served to perform further essential tests and trials in anticipation of a descent to the Moon, and soon after, the crew of Apollo 10 also reached to lunar orbit (Zou et al., 2015, Lowman, 1969). On July 16, 1969, Apollo 11 departed Earth in the Saturn V launch vehicle, shown in Figure 2. The crew consisted of Neil Armstrong, Michael Collins, and Edwin "Buzz" Aldrin (Figure 3). After one orbit around the Earth and a four-day journey to the Moon, the command module entered lunar orbit and on July 20, the LEM Eagle, carrying Armstrong and Aldrin, separated from the command module and began the approach to the Moon. Neil Armstrong was the first man to set foot on the Moon, followed by Edwin Aldrin. On the same day, docking back to the command module took place and Apollo 11 began the return to Earth with 21 kilograms of lunar rock samples (Harland, 2010).



*Figure 2: Saturn V lift-off(NASA-Press-Kit)*



*Figure 3: Apollo 11 crew: Neil Armstrong (left), Michael Collins (centre) and Buzz Aldrin (right). Credit (NASA-Press-Kit)*

In the same year Apollo 12 also successfully landed on the Moon (Warner, 1970). The 1970 Apollo 13 mission suffered a major setback when an explosion occurred preventing the astronauts from descending to the Moon and forcing them to make an emergency return to Earth (Lovell and Kluger, 2006). Three missions (Apollos 14, 15 and 16) followed, allowing further exploration of the Moon by astronauts. In December 1972 the final mission of the program, Apollo 17, was launched (Jones, 1995). A total of six Apollo missions launched between 1969 and 1972, and twelve astronauts walked on the Moon.

After the Apollo Program was successfully completed, the United States government began to reduce funding to NASA, which led to the cancellation of further lunar missions. Instead, NASA's priorities changed in the 1970s and 1980s, when the agency focused on the Space Shuttle development and solar system exploration with automated probes. In addition, with the fall of the Soviet Union and the subsequent end of the Cold War in 1991, the political and financial drive to conquer space decreased significantly, leading to a further loss of interest in lunar missions (Thomas et al., 2013). This was until Japan's Hiten mission of 1990 was launched with the goal of exploring the Moon (Uesugi et al., 1991). In 1994, NASA's Clementine mission proved the presence of water in the lunar polar regions, and since then interest in lunar exploration has increased again (Nozette et al., 1994).

Many nations have since developed space programs. Easier access to space has been made possible by the creation of international organizations such as the European Space Agency (ESA). It enabled countries to collaborate and share resources and advanced technologies at a faster rate. Following this, new space powers such as China, India and Japan have emerged. In 2003, ESA launched the Small Missions for Advanced Research in Technology-1 (SMART-1) probe, the first European probe for the exploration of the Moon. SMART-1 carried out a complete reconnaissance of the lunar surface, including X-ray surveillance, and ended on September 3, 2006 with a controlled crash on the surface (Foing et al., 2006). The Japan Aerospace Exploration Agency (JAXA) launched its second lunar mission, the Selenological and Engineering Explorer (SELENE) in 2007, which made a significant contribution in mapping gravitational anomalies on the Moon. (Kato et al., 2008).

China is one of the most recent nations to become space-faring, being actively involved in space exploration only from the beginning of this new century. The Chinese National Space Administration (CNSA) made significant achievements in Moon exploration, starting the Chinese Lunar Exploration Program (CLEP). The CLEP consists of three stages involving a series of lunar orbiters, landers, rovers, and missions to collect samples from the Moon's surface, launched aboard Long March launchers (Gibson, 1998). The Chang'e 1 orbiter was launched on October 24, 2007 (Ouyang et al., 2010). On October 1, 2010, the Chang'e 2 orbiter was launched (Li et al., 2018), and three years later the Chang'e 3 mission landed the Yutu rover near the Sinus Iridum region in 2013, the first object to make a lunar soft landing since the Luna 24 probe in 1976 (Basilevsky et al., 2015). The Chang'e 4 lander, shown in Figure 4 was destined for exploration of the far side of the Moon, where it successfully landed on January 3, 2019, near the Von Kármán crater (Jia et al., 2018). The lander released the Yutu-2 rover onto the surface (Lai et al., 2020) which is shown in Figure 4. The Chang'e 5 mission, launched in November 2020 and consisting of a lander, orbiter and ascent module, successfully picked up and brought back to Earth about 2 kilograms of lunar rocks (Lin et al., 2022). In the future, the CNSA intends to explore the lunar polar regions, aimed specifically at the construction of space stations, lunar research stations, and lunar bases, (Zheng et al., 2008, Xu et al., 2020, Wu et al., 2019). In preparation for the upcoming lunar activities, the CNSA proposed the

establishment of the joint International Lunar Research Station (ILRS) in 2016 (Jones et al., 2021, Jianlan, 2019).

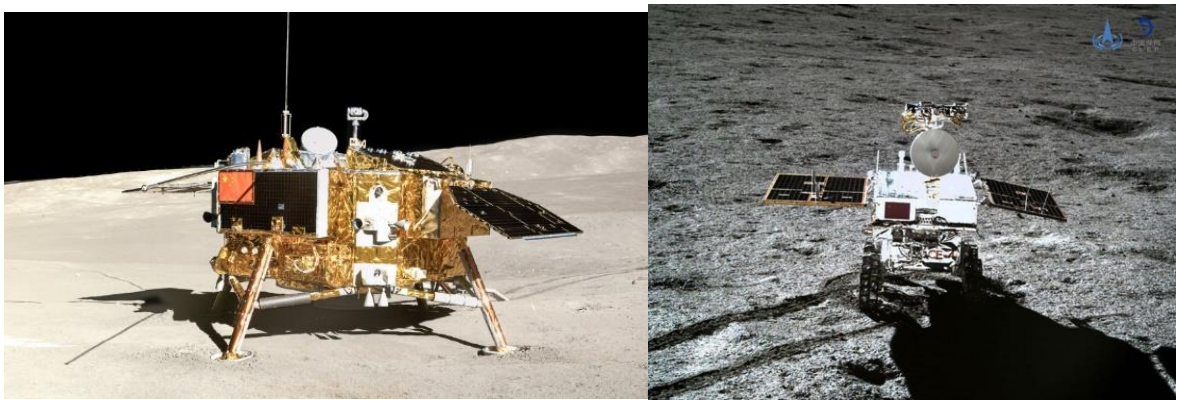


*Figure 4: Chang'e 4 lander on the Moon (left); The Yutu 2 rover on the Moon's far side, as imaged by the Chang'e 4 lander (right). Credit CNSA*

The Indian Space Research Organization (ISRO) marked India's first steps into space with the Chandrayaan program. The lunar orbiter Chandrayaan-1 was launched on October 22, 2008 (Goswami and Annadurai, 2008). The two-year planned mission was to create a three-dimensional lunar map, conduct chemical and mineralogical mapping of the surface, and release the Moon impact probe that crashed into the surface on November 14. Among the various scientific discoveries, the probe confirmed the presence of water molecules in the lunar soil (Pieters et al., 2009). The follow-up mission Chandrayaan-2 was launched on July 22, 2019, and consisted of an orbiter, the Vikram lander, and the Pragyan rover (Khati and Sharma). However, the descent of the lander and rover failed due to a malfunction.

The Israeli probe Beresheet, a lander technology demonstrator that was intended to be the first privately funded probe to reach the Moon, failed to land on April 11, 2019 (Shyldkrot et al., 2019, Gibney, 2019).

Returning to the USA, in 2009, NASA launched the Lunar Reconnaissance Orbiter (LRO) mission, which created a high-resolution three-dimensional map of the surface and delivered the Lunar Crater Observing and Sensing Satellite (LCROSS) impact probe (Vondrak et al., 2010). The impact, which occurred on October 9, 2009 near Cabeus crater, allowed for the study of debris raised by the collision, and provided further evidence of the presence of water (Mitrofanov et al., 2012). In 2011, the Gravity Recovery and Interior Laboratory (GRAIL) mission put two probes into lunar orbit to measure lunar gravitational field anomalies and gather information about the composition of the crust and mantle (Zuber et al., 2013).

## 3.3 Current State of Governmental Lunar Exploration

As demonstrated, in the last few decades there has been a resurgence in lunar exploration in pursuit of scientific and technological advancement, with a sustained human presence on the Moon before the end of this decade being a highly desirable outcome. One of the main objectives is to utilize a lunar outpost as the staging point for exploring our solar system with the insights gained from lunar explorations to date (Pickrell, 2022). Although several lunar missions are planned in the coming years by countries such as Japan, South Korea, Russia, India, the United Arab Emirates and the United States (Pickrell, 2022), this section reports on current programs with the intent of returning humans to the Moon.

### 3.3.1 Artemis

Artemis is a lunar exploration program by NASA with the objective of the civil exploration of the Moon through international partnerships and collaboration with private industry. One of the primary aims of Artemis is to establish a reliable and sustainable process for the extraction and use of space resources in accordance with the Outer Space Treaty (OST). Key to the Artemis program is NASA's Exploration Strategy outlining the transition from government to commercial providers of access to Low Earth Orbit (LEO), and the successful establishment of partnerships with global economies that will aid in the long-term robotic and human exploration of cislunar space and the Moon. The successful implementation of these two domains, as well as the development of new technologies and capabilities will be applied to the ultimate goal which is to support the first human flight to Mars.

The Artemis program currently consists of four missions: Artemis I, II, III, and IV, with many more planned, but not yet finalized. These are all part of NASA's Moon to Mars Strategy (NASA, 2020).

- **Artemis I - Uncrewed Flight Test of the SLS and Orion**

Artemis I demonstrated the first successful integrated uncrewed flight test of the Space Launch System (SLS) and the Orion spacecraft on 16 November 2022 to validate the systems' performance in deep space, and Orion's thermal resilience to Earth-return speeds (see Figure 5). The SLS, a super heavy-lift expendable launch vehicle developed by NASA is designed to launch the crewed Orion spacecraft on a trans-lunar trajectory.

- **Artemis II - Crewed Flight Test**

The crewed flight will validate the life support systems, critical subsystems, and flight controls in a docking and proximity operations test to be launched in 2024 (Weber et al., 2021). In addition, 13 CubeSats will be deployed, engaging universities and companies in lunar exploration (NASA, 2020).

- **Artemis III – Crewed Flight Landing First Woman and Next Man on the Moon in 2025**

Artemis III will land the first woman and next man on the Moon. This mission will see astronauts aboard Orion dock in 2025 with a Human Landing System (HLS) vehicle, to be designed and built by SpaceX (NASA-Artemis3, 2023). From there, two astronauts will make their descent to the lunar South Pole where they will spend about 6.5 days on the surface, living inside the HLS crew cabin that will then return them to lunar orbit to dock with Orion (Weber et al., 2021).

- **Artemis IV and beyond – Moon**

It is important to note that the Moon is a "stepping stone for the human exploration of Mars". As such the core element of the sustained lunar presence will be the expansion of infrastructure at the Gateway (NASA), a staging point for human and robotic missions to the Moon. It will be a habitable space station orbiting the Moon, and is crucial to sustainable lunar exploration (NASA-Gateway).

Notice that while Artemis I to III provide a blueprint detailing how to reach the Moon, subsequent missions (Artemis IV and beyond) will determine how a sustainable human presence will be established on the lunar surface. Moreover, it must be considered that all these practical actions are intended to be regulated by a system of international laws and treaties. In particular, the Artemis Accords, which as of December 2022 have been signed by 23 states and one territory, already outline the shape that international cooperation for scientific exploration and commercial

exploitation of the Moon will take in the near future, through a series of bilateral framework agreements.

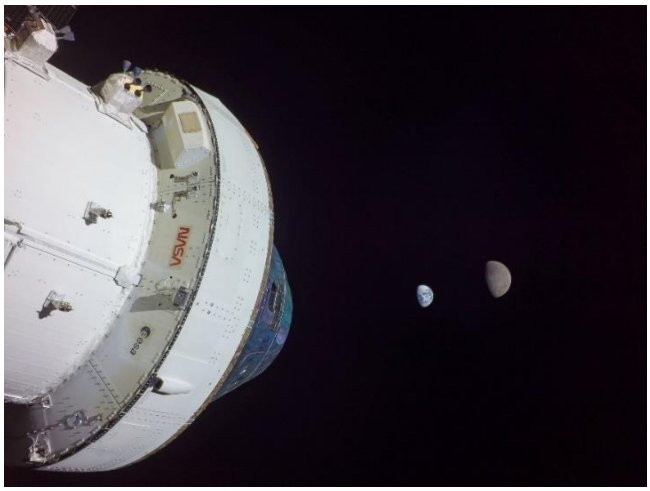



*Figure 5: Artemis I. Credit NASA*

### 3.3.2 International Lunar Research Station

On March 9, 2021, the CNSA and the Russian Space Agency (ROSCOSMOS) signed a Memorandum of Understanding (MoU) for the joint construction of an autonomous permanent research base on the Moon (Wu, 2023) opening negotiations with international partners.

The program, called International Lunar Research Station (ILRS), is a wide range of scientific investigations aiming to advance our understanding of the Moon, its environment, and resources, as well as to demonstrate new technologies and capabilities for future human and robotic exploration.

The ILRS is connected to the previous CLEP which sees a series of missions like Chang'e-6 to 8 which aim to increase knowledge of the lunar south pole as the last phase (Xu lin, 2022). This program is designed to attract global partners and, through a facility built on the Moon, aims to undertake a wide variety of scientific research in line with the ILRS research areas of establishing a geological survey of the Moon, lunar based ISRU, science experiments and astronomy, as well as the overall observation of the interactions between the Sun, Moon and Earth (Xu lin, 2022).

The ILRS construction is scheduled in three parts: reconnaissance, construction, and utilization. The reconnaissance phase is scheduled to occur before 2026 with the Chang'e-6 mission, which aims to collect and return samples from the far side of the Moon. Chang'e-7 will perform an in-depth investigation of the characteristics of the lunar south pole and assessment of its suitability for a human base camp (Hu et al., 2022). Also at this stage, ROSCOSMOS in cooperation with international partners will launch the Luna 24, 25, and 26 missions (Malakhov et al., 2015) (Litvak et al., 2022), (Shakun et al., 2017) (Bradshaw, 2017).

The second phase of the ILRS, the construction stage, is scheduled to be completed before 2035, first through the Chang'e-8 and Luna 28 missions that will detect the physical and geological features of the proposed building site, then by performing the first in-situ experimentation in both the process of extracting rare gases and the creation of small Earth ecosystems on the Moon. From 2031 to 2035, another five missions, ILRS 1-5 are scheduled to complete the in orbit and surface facilities for energy, communication, and transportation services as well as for research, exploration,

scientific experimentation and verification of technological principles to be applied in following explorations, such as lunar experiments and sample collection.

Finally, after 2036, with the utilization phase, the ILRS base would be used for scientific and technical verification purposes through a series of crewed lunar missions. This platform will allow the different partners involved to carry out a long term, constant observation and exploration of the Moon.

The facilities of the ILRS are the following:

- Cislunar Transportation Facility
- Long-term Support Facility on Lunar Surface
- Lunar Transportation and Operation Facility
- Scientific Facility
- Ground Support and Application Facility

### 3.3.3 ESA Moon Village

Moon Village was formally introduced under the new ESA leadership appointed in 2015. It was considered an understanding, where ESA would lead in the role of initiator and coordinator, leading discussions with other agencies and with commercial stakeholders to define potential contributions to realize their vision.

The driver behind the concept was that without the ISS there would no longer be a grand international scientific project in space. Moon Village was proposed as a potential successor to the ISS. Moon Village became the first and is still considered the only idea for the post-ISS era. Mounting misconceptions, lack of funding by ESA, and a change in leadership at ESA may have been contributing factors for the concept to lose traction, and it was eventually quietly abandoned in 2020. Following the appointment of new ESA Director General Josef Aschber in 2021 a new strategy was introduced in 2021, the Terrae Novae 2030+ Strategy Roadmap (Wolny, 2017).

### 3.3.4 Terrae Novae (ESA)

The document outlines a range of goals that have been identified as achievable in the Terrae Novae 2030+ strategy of intent, such as creating a post-ISS presence of Europeans in LEO, followed by the first Europeans to step on the lunar surface by 2030, ultimately enabling a European contribution to the first human missions to Mars (ESA, 2022).

ESA states (ESA, 2022) that it aims to achieve those objectives through a shift from an inhouse procurement approach to one that relies on integration of commercial actors in the private sector through the purchase of end-to-end services. However, the program is not completely impermeable to international cooperation.

ESA is a signatory of the Artemis Accords and will collaborate on the Artemis program, and it is a potential contributor to the ILRS. Terrae Novae, as a program, is still theoretical and would occur too far in the future to be of relevance to this project. For these reasons, it will not be discussed further, despite ESA being a major entity in the space sector.

## 3.4 Commercial Lunar Programs

In addition to government agency efforts, there are commercial programs that aim to establish a continued presence on the Moon, providing services and infrastructure for future space travel and the development of advanced technologies. This has the potential to create economic and business opportunities related to lunar exploration. The practice of commercial companies and private

entities facilitating space flight and exploration is known as the New Space venture (Kodheli et al., 2020).

The commercial space sector is rapidly expanding due to increasing interest and investment. With institutions such as Goldman-Sachs estimating the New Space market as having a value of greater than $3 Trillion by 2040 (Crane et al., 2020), many new players are entering this quickly growing sector. NASA is already benefiting from strong commercial collaboration, with contracts established with private companies already. Their objective is towards a long term sustained presence on the moon.

To date, Blue Origin, Dynetics, and SpaceX have secured contracts to begin refining Human Landing System (HLS) designs for the Artemis program (Mahoney, 2020). For a full list of commercial projects past and present refer to Appendix A.

It is worth noting that there is an interdependency between NASA and the commercial sector though. This is due to the financial flexibility of corporations, or an accommodating philosophy of learning through failure. Something that stands in stark contrast to the strict budgetary constraints governmental organisations are bound to.

Some of the most significant recent and current projects are discussed below.

- **Commercial Lunar Payload Services (CLPS)**

Under this USD $2.3bn initiative NASA is working with an array of American companies to deliver science and technology to the lunar surface. Their involvement sees the delivery of payloads for NASA, including payload integration and operations, launching from Earth and landing on the surface of the Moon.

Primary precipitants of NASA's Commercial Lunar Payload Services are participants of Lunar XPRIZE, a competition launched by Google in 2007 worth $30 million in prizes. The terms included for independent commercial teams to undertake the mission of "Landing a robot on the surface of the moon, travelling 500 meters over the lunar surface, and sending images and data back to earth." (Barton et al., 2018).

Whilst the main prize went unclaimed in 2018, the initiative sparked competition in the commercial lunar space sector with major teams continuing lunar rover work, ventures including iSpace, Astrobotics, SpaceIL, and Moon Express.

- **dearMoon Project – SpaceX (2023)**

The dearMoon project aims to be the first civilian mission to the Moon, conceived by Japanese billionaire Yusaku Maezawa. Using a SpaceX starship rocket, the mission is slated to launch in 2023, undertaking a circumlunar trajectory. The trip includes a crew of eight, plus mission specialists. This is the first planned entirely private mission to the Moon in the domain of space tourism (Howell, 2022).

- **Hakuto-R 1 – iSpace (2022)**

iSpace is designing landers and rovers in order to compete for commercial contracts associated with lunar exploration and transportation. The Hakuto-R 1 mission aims to be the first successful commercial lander on the Moon (Figure 6). The mission contains an iSpace payload, the United Arab Emirates Rashid Rover, and JAXA's SORA-Q rover.

The Hakuto-R 1 mission used a Falcon 9 launch vehicle on 11 December 2022. The mission will spend a month in deep space, followed by two weeks in orbit before a planned soft landing on the Atlas crater in March 2023 (iSpace, 2023).



*Figure 6: iSpace Hakuto R-1 lunar lander. Credit iSpace*

- **Mission 1 – Astrobotic (2023)**

In the first quarter of 2023, Astrobotic's Mission One will carry 28 payloads to the Moon. Of these payloads, 14 are commercial and 14 are part of NASA's Commercial Lunar Payload Services. The launch vehicle used will be United Launch Alliance's Vulcan Centaur rocket.

In line with Artemis II, the Peregrine 1 mission was selected for the CLPS initiative, in which commercial partner Astrobotic provides NASA with the lander, illustrated in Figure 7. To increase the scientific value of the mission and align with the Artemis activities, the original landing site was redirected and payload adjusted (NASA, 2023b).

The goal of Mission 1 is to establish Astrobotics as a commercial provider of lunar orbit and lunar surface payload delivery. It will also undertake the scientific objectives of studying the lunar exosphere, radiation, lunar regolith properties, and lunar magnetic properties (Williams, 2022b).

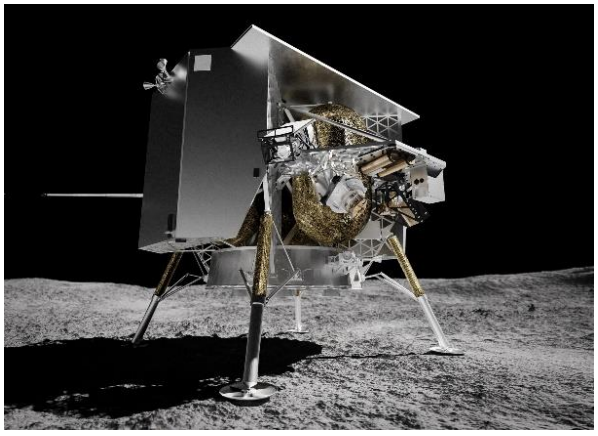


*Figure 7: Astrobotic Peregrine Lunar Lander. Credit Astrobotic*

- **Nova C – Intuitive Machines (2023)**

Nova-C is a lander vehicle designed to deliver small payloads to the surface of the Moon. The mission will launch in March 2023 using a Falcon 9 launch vehicle. The Nova-C lander, visible in Figure 8, payloads have a focus on raw material processing on the Moon. While originally planned to land in the Vallis Schroteri region of the Moon, as NASA's objectives have changed, so has Nova-C's

landing location. Currently, the landing area is at the Moon's south pole. Nova-C is planned to carry eleven payloads, with five as part of NASA's Commercial Lunar Payload Services (Williams, 2022a).

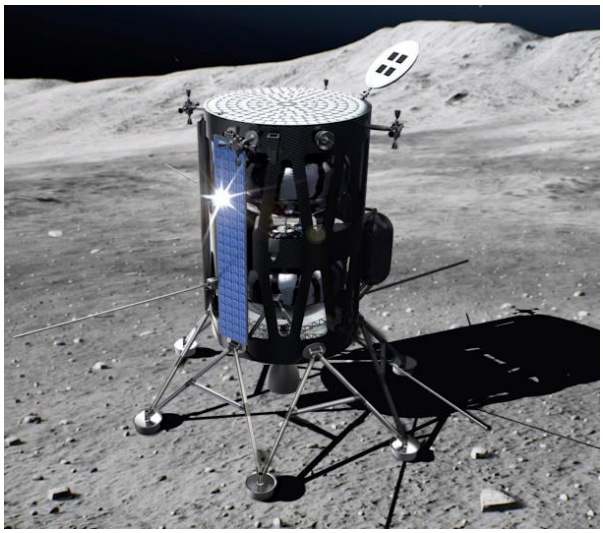


*Figure 8: Intuitive Machines 1 Nova C lunar lander. Credit NASA*

## 3.5 Comparison of Artemis and ILRS

In the previous sections, three lunar exploration programs have been outlined. Artemis and the ILRS will occur close enough to the present that they will be relevant for the purposes of this report, while Terrae Novae is too immature, so it will not be considered. A comparison between Artemis and the ILRS is provided below, describing their similarities and differences.

In summary, both programs aim to explore the Moon, and share a common urgency in achieving a human return to the Moon in the most feasible, fastest way possible. However, the focus and specific objectives are different: Artemis focuses on returning humans to the moon with the Artemis Base Camp (ABC), providing a structured outline of multiple sequential objectives to achieve this in the near- to mid-term. Furthermore, it provides a gateway for deep space exploration and human missions to Mars. ILRS focuses primarily on conducting scientific experiments and exploring mineral resources on the Moon, and using the station for human presence only in the final phase. The potential suitable areas on the lunar surface for a sustainable settlement are as now found in the lunar south pole, and the areas where conditions align are contested and limited. It may be assumed that once the first party reach these areas, a buffer zone such as the one outlined in section 11 of the Artemis Accords may be created citing safety concerns, thus preventing the other programmes or actors from accessing vital resources and thus greatly lowering the program success or power.

It is thought that Artemis, until the Gateway component is active, is solely focussed on identifying landing sites to get the first people back on the lunar surface. It thus relies on information from previous or ongoing programs to infer suitable landing sites. The establishment of a long-term lunar base is not mentioned.

The combined Chinese and Russian approach through ILRS in turn have a more systematic approach. The missions are sequential, each stating clear objectives. This approach appears more methodological and the areas selected are not exclusively considered as potential landing sites. The sites have also been deemed suitable for the establishment of a lunar base.

A further crucial aspect of comparison between the two programs is the level of international cooperation. A lack of cooperation between the main actors of Artemis, and ILRS, namely the USA

for one and China and Russia may be due to restrictions imposed on NASA by the Wolf Amendment. Given the harsh environment on the lunar surface, there may arise a need for common protocols to allow for international collaboration to address and support arising safety concerns. International partnerships, including commercial contracts, to achieve the Artemis objectives have been established, leading to the USA-led Artemis Accords. Whereas for China and Russia, not bound by the Artemis Accords, the open invite and encouragement to unconditional international scientific collaboration for the ILRS program is evident. They are actively encouraging countries to include their scientific payloads on missions as of now signing a version of accords before, and incentivising them to join, possibly through the leverage created through the belt and road initiative ran throughout developing countries by China.

International partnership known for this program are with Thailand and Iran. However, while they strongly emphasize preserving scientific integrity, there is a certain lack of transparency in relation to updates on timeline and budget. To date, for ILRS, a comprehensive feasibility study identifying landing sites and potential human settlement sites on the south pole of the Moon has been conducted (Hu et al., 2022). However, the ILRS program offers no details on the establishment of sustainable human lunar settlement.

## 3.6 Conclusions

This section has explored the history of spaceflight and the status of current human lunar missions, concluding that there are many contributing factors that make these efforts possible. Among them are technological capabilities, economic feasibility, and international and legal considerations as many nations seek to expand to the Moon and beyond. All of these factors together must be addressed to enable a long-term sustainable human presence on the Moon and further explorations.

Potential for major geopolitical concerns may arise out of competing interests between Artemis and ILRS. First of all, the potential geopolitical driver promoting space currently may include a challenge of leadership in the lunar environment. In fact, leadership in space capability is often connected to global political and military power on Earth. Another highlight is the large amount of commercial growth and interest in space ventures. This adds credence to the potential economic value of establishing a sustainable presence on the Moon, but it emphasizes the lack of clarity in terms of international laws on the commercial exploitation of space. This project addresses these legal issues in the Legal Framework section by proposing a mechanism to regulate government and commercial activities in cislunar space and on the Moon.

From the considerations presented, it is clear how the technological and economic impacts connect with the development of a human presence on the Moon. Moreover, it is of primary importance not to forget that all these programs act on a new land, and require the interaction of different international participants, each of which pursues its own goal. Therefore, during this research activity it has been decided to focus on how to achieve all these objectives in a sustainable way, both from a technological point of view and in legal terms. In particular, Resource Utilization illustrates possible ways to evolve from current programs, and utilize the resources present on the Moon for a sustainable human presence. In Legal Framework, a new legal framework is presented, which considers the management of international conflicts and all the possible implications of the fact different nations are competing to achieve the same goal. It presents an updated treaty for the use of cislunar space, the Moon, and resources, as well as a supranational system to ensure compliance with these guidelines.

# 4 Resource Utilization

## 4.1 Introduction

As presented in Evolution of Lunar Exploration, a series of programs targeting the exploitation of the Moon, and its future settlement, have been developed by various nations and private companies. However, as the Moon represents a land belonging to no nation, the question of how to make such an activity sustainable, in both a physical and geopolitical sense, has arisen.

While sustainability has many definitions – such as environmental sustainability – which are pertinent to this report, the one that will be focused on is the ability to maintain a continuous long-term human presence on the Moon. Factors required to achieve sustainability include resources, economic, legal, cultural, political, ethical, and scientific considerations. After extensive research and due to its criticality for a long-term human presence on the Moon we selected resources as the main focus of this section of the study.

The concept of resources is a very broad one, having both physical and abstract meanings, particularly in the context of space exploration. For this reason, this section of the report focuses primarily on a handful of resources which relate most strongly to the physical sustainability of a lunar settlement.

For this research, the three selected aspects of lunar settlement development are: power, in-situ resource utilization (ISRU), and human habitation. Many other resources, supporting utilities, and components of the economy which were considered, but ultimately not included in this study, include communication, human adaptability, remote sensing, medicine and remote health care, reusability of launchers and landers, tourism, artificial intelligence, and human genome editing.

Since the achievability of long-term human presence is directly correlated with the technology and research available, this team project addresses the challenge of sustainability by considering the following roadmap: near-term (until 2030), mid-term (until 2040) and long-term (until 2050). This report presents the challenges in sustained human settlement on the Moon for each of the selected resources, the current technologies available for overcoming those challenges, and recommendations for the future of human presence on the lunar surface. These recommendations are not intended as a statement, so much as they are suggestions. We acknowledge that the version of events we present in this roadmap and in our recommendations is a single scenario, which has many alternatives that can be explored by stakeholders.

## 4.2 Roadmap

The roadmap for the long-term presence of humans on the Moon, is illustrated in Figure 9.

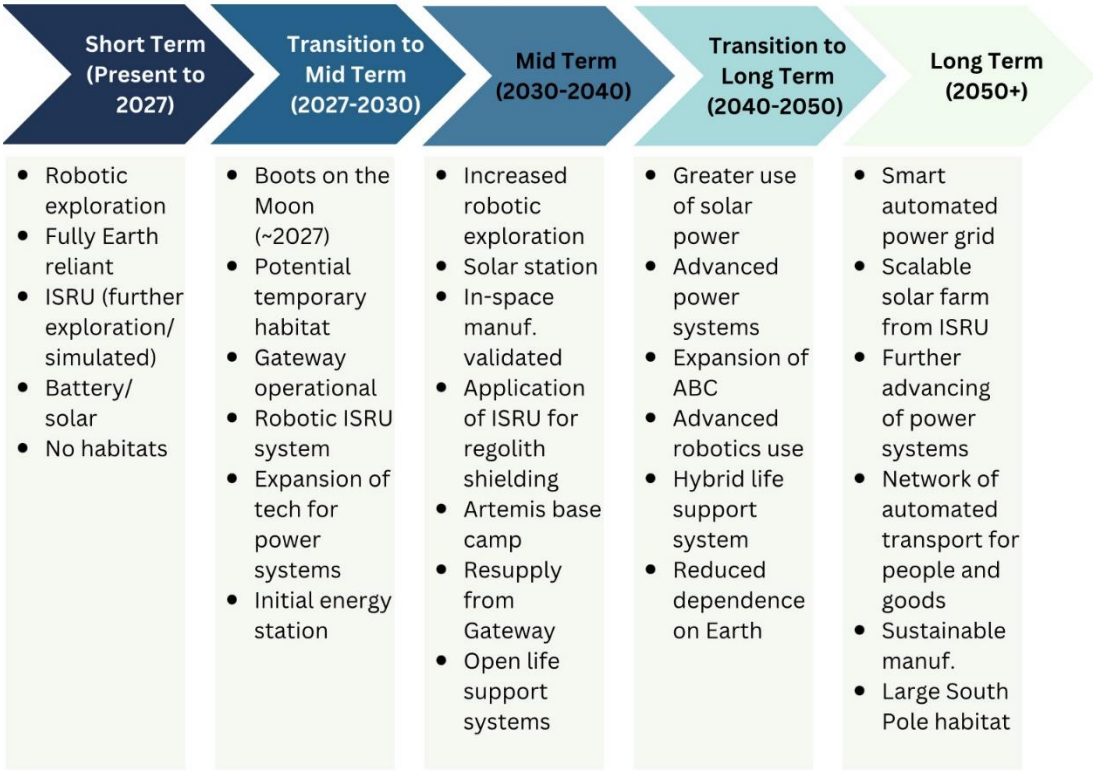



*Figure 9: A recommended scenario for the development of human presence on the Moon.*

## 4.3 Power

### 4.3.1 Power Requirements for a Lunar Habitat

To achieve long-term human presence on the Moon, a power supply system available for both day and night is essential for human life support, mobility, and research capability. The infrastructure needs to be progressively scalable. Near-term missions utilizing only robotic exploration with landers and rovers should have minimal external power requirements. The power requirements for short- to mid-term are instead expected to range from 25 kilowatts (kW) to hundreds of kW for scientific activities (Waldron, 1990, Palos et al., 2020). Finally, when transitioning to a fully operational settlement with ISRU production and closed-loop life support, power levels of megawatts will be required (Palos et al., 2020, Waldron, 1990). In this section, the focus is on the three major components of a power system: power generation, energy storage, and power distribution. The state-of-art of related technologies will be reviewed in the next few sections, and the challenges and recommendations for short-, mid-, and long-term mission usage will be discussed.

### 4.3.2 Power Generation

Currently, various energy supply solutions, such as photovoltaic arrays (PVAs), radioisotope thermoelectric generators (RTGs) and fission power systems (FPSs) have been proposed for sustained lunar presence. (Bermudez-Garcia et al., 2021, O'Brien et al., 2008, Elliott et al., 2006) These are detailed in the following sections.

- **Photovoltaic Arrays**

For the near-term, the most promising candidates for power generation are photovoltaic cells and regenerative fuel cells, providing a clean and plentiful energy resource on the Moon. Solar cells are

flight qualified and have been used in most prior planetary missions as the primary source of power. There are still four main challenges for photovoltaic power generation on the Moon that must be overcome:

- long term degradation and sudden decrease of solar cell efficiency under extreme proton radiation and solar particle events,
- lower efficiency of the solar cells at relatively high operational temperatures,
- no direct sunlight for photovoltaic power generation during the lunar night, and
- blocking of light on photovoltaic cells by lunar dust (Girish and Aranya, 2012)

There are efforts to address these issues by redesigning the cells and developing new materials. Cooling techniques and concentrator photovoltaics (CPV) arrays currently under development are increasing the efficiency and survivability of solar cells (Sharaf et al., 2022, Bermudez-Garcia et al., 2021, Khamooshi et al., 2014). The areas known as the peaks of eternal light near the lunar south pole receive almost constant sunlight, which offers the prospect of generating near-continuous solar power. Locating solar panels at the south pole near the Shackleton crater will provide power for mining water and other volatiles in the nearby permanently dark regions. Using vertical solar arrays or solar arrays on high towers will offer a solution to receive more sunlight.

To enable mid- to long-term human missions to the Moon, more research about new materials and designs for a photovoltaic power generation system with higher efficiency are needed. Even though research has been carried out proposing electrostatic, ultrasonic or brushes for dust removal on solar panels, they are still in very early stages (Shahmoradi et al., 2020). In the long term, with further development of solar cells, it may be possible to place space solar powered satellites in a lunar sun-synchronous orbit and produce electricity which could be converted and transmitted using Microwave Power Transmission (MPT) or Laser Power Transmission (LPT) to a receiving site on the Moon, and potentially in the future even Mars missions (Baraskar et al., 2022). Early research shows that it might be possible to use the regolith to make solar cells by ISRU, as silicon is one of the most abundant elements on Moon (Ignatiev et al., 2004, Blue Origin, 2023a)

- **Radioisotope Power System**

Radioisotope power systems (RPS) do not use nuclear fission or fusion, instead they convert heat from the natural radioactive decay of plutonium-238 (238Pu), a radioactive isotope, into electrical power. There are two types of RPS; radioisotope heater units (RHU), and radioisotope thermoelectric generators (RTGs) (O'Brien et al., 2008, Lange and Carroll, 2008). RTGs are space-qualified and have contributed to many historic space missions. The most current RTG model, the multi-mission radioisotope thermoelectric generator (MMRTG), has been used by NASA on the Mars 2020 Perseverance rover mission – providing around 110 Watts when freshly fueled – and will be used in the Dragonfly rotorcraft, an astrobiology mission which will explore Titan (NASA, 2020). Radioisotope power sources can provide both thermal control and electrical power, and have a long operational life. With the compact design, no moving parts, and minimal radiation effects on equipment or personnel, RTGs can withstand the harsh environments, require minimal maintenance and can be a reliable energy source for remote operations. It is proposed to use RTGs in the short- to mid-term, especially for the early robotic missions. RTGs could also be used to power transportation such as rovers during two- to four-person crewed missions in the mid-term. However, RTGs need to be built on Earth at significant cost and launched to the Moon, and the amount of 238Pu is limited, which is not sustainable in the longer term.

- **Fission power systems**

Fission power systems (FPS) rely on a sustained fission reaction of uranium-253 (235U) and will offer the potential supply of electric power from kilowatts to megawatts. There was a successful ground test of a FPS by NASA, which produced a power output between 1 to 10 kW (Mason, 2018). Recently, NASA has selected three design concept proposals and is working to design an FPS that would provide at least 40 kW of power, which is enough to continuously run thirty households for ten years (Harbaugh, 2021). However, there are currently no space-qualified FPS due to a lack of technological knowledge. In particular, there is a need to develop new reactor fuels, structural materials with reduced mass and long operational life. So, this technology is recommended for further development to address power needs in the mid- to long-term.

### 4.3.3 Energy Storage

For the purpose of the study, we have considered the possibility of a solar powered Moon base, which implies that there will be issues with intermittent solar power generation, due to the day-night cycle. However, this problem is not new, since the same considerations have been encountered in the past, for example on the ISS.

The batteries used in ISS are for backup energy in emergencies, and to power the station when the regular dark periods, approximately 45 minutes long, prevent solar power generation. This principle is used as a basis to extrapolate the necessity and requirements for an energy storage station on the Moon, where dark periods can be up to fourteen days long at the equator. In the area of interest near the poles, “peaks of eternal light” exist where the lunar surface experiences near-constant illumination, reducing the length of darkness. The length of these dark periods means a greater ability to store power is needed than is currently used on the ISS.

In emergency cases and for backup energy, the ISS uses banks of Lithium-ion batteries to store power (Garcia, 2021). However, multiple other types of energy storage exist. There are four current varieties of rechargeable batteries that could be used on a lunar base in the near- to mid-term:

- Sodium-ion battery (Na-ion)
- Nickel-hydrogen batteries (Ni-$H_2$)
- Hydrogen Fuel cell
- Lithium-ion battery (Li-ion)

The primary advantage of each of these types of batteries is that we can find the materials to make them on the Moon (see In-Space Manufacturing section). It is worth noting that Na is more common than both Ni and Li, and the making of Ni batteries generates fewer contaminants than the making of Li batteries.

In the case of ISRU for hydrogen, a critical aspect is the fact that hydrogen storage presents a safety challenge, specifically because it is combustible. However, it has already been stored and used safely in the Space Shuttle program as fuel, and has a history of successful usage in the Apollo missions. With the correct infrastructure, safe storage could be possible on the Moon. This would make it possible to refuel spacecraft on the Moon, and liquid hydrogen would be available for other applicable processes like enriching the air, recombining with oxygen for potable water, and energy storage (Heiney, 2021).

It is this report’s recommendation that the near-term settlement of the moon should utilize Li-ion batteries for backup power and storage until the first solar station can be installed. Beyond this,

backup capacity should continue to expand using Li-ion batteries until water processing (see Life Support Systems section) enables the production of hydrogen fuel cells, which would have the advantage of storage infrastructure for hydrogen already being present on the surface. In the future, it will be necessary to have a dynamic and inexpensive method of storing energy. Robotic workers will need some means of allowing continuous operation in permanently shadowed regions (PSRs), where solar power will not be available. However, relying solely on Li-ion batteries sent from Earth is not scalable or sustainable, so a battery-manufacturing plant on the moon may need to be constructed to facilitate future activities.

This proposed facility would utilize Na, $H_2$, and Ni, which will be readily available in-situ by this time. Different types of energy storage will be in demand: Na-ion batteries are cheaper to produce, and the materials are easier to find, however they have lower energy density, lower thermal stability, and life cycle compared with Ni-H2 (de la Llave et al., 2016, Nakahara et al., 2010). Nonetheless, the flexible and lightweight battery technology will put them as the first developed approach to phase out Li-ion based batteries.

These solutions facilitate a robust energy storage system that, combined with efficient power distribution, will allow humans to use available technologies to their fullest potential to overcome the challenges of the lunar environment. As long as capacity and requirements continue to expand it will be necessary for mission sustainability to manufacture energy generators and energy storage as part of ISRU.

### 4.3.4 Power Management and Distribution

Power sources and power storage are connected to loads by the power management and distribution (PMAD) system (Kenny and Coleman, 1991). These are systems that are already used on Earth, but adapting them to the challenges of a space environment, and the need to automate processes, will drive technology development to provide the foundational infrastructure for power supply to a sustainable lunar habitat and economy (Khan et al., 2006).

Systems that already exist and operate in space have self-contained power management, while Earth-based PMAD is mature for large power grids over entire nations. Initial efforts on the Moon will be self-contained such as those already planned in the Artemis missions, such as the lunar lander, robotic rovers, the first habitation module, and pressurized and unpressurized rovers (Kessler et al., 2022). These will be fully self-contained systems with integrated power source, storage, distribution, and management. As the settlement grows and expands, vertical solar panel arrays will be deployed to provide power to multiple systems, this will be the start of the micro grid (MG), a smaller version of power grids on Earth (Smith, 2021a, Scott, 2022).These vertical solar arrays, best suited to the polar regions of the Moon, will be the transition point to micro grid systems on the Moon.

This mid-term stage will create micro grids connecting a local power source to what is needed around it (Saha et al., 2021). Power will be supplied to multiple habitats in a closed system that draws power from a central source. Rovers will be able to plug in to charge, then transport people and things to other locations that may be on a separate micro grid, like a lab facility near a geological place of interest, or ISRU collection and water-production stations (Scott, 2022). The technology development challenge that must be focused on is creating autonomous control systems that are smart and modular so they can handle critical and shifting loads (Csank et al., 2019). The MGs will

eventually connect to other MGs as infrastructure is built up (Saha et al., 2021). These systems need to be tested and validated to be safe, autonomous, and efficient for the best use of power sources, particularly those using ISRU and supporting human habitats.

Transitioning to larger power loads and bigger grids will be triggered with the combining of MGs and use of nuclear fission reactor, with solar power used to power foundational infrastructure for power management and distribution (Harbaugh, 2021). The focus will shift to power for ISRU. New sources and storage can dynamically be added to the scalable MGs.

In the long-term the PMAD will need to sustain a grid large enough to support the power needs of multiple habitats, as well as ISRU systems producing critical water and oxygen, initial tourist activities, and early solar panel production (Kristmann et al., 2022, Scott, 2022). Continued development will be needed on wireless transmission technology and the systems required to integrate it into the power grid. This will enable a centralized location to receive scalable power from space-based solar panels which can also supply power to Earth or other cislunar regions. The management and distribution technology development done on the Moon will also be applicable on Earth. This recommended long term PMAD vision will provide an automated flexible grid system to power the lunar infrastructure with available power generation sources and storage technologies that can extend to the full surface of the Moon and support the many human habitats that will be created for human activities and growth.

## 4.4 In-Situ Resource Utilization

To enable sustainable infrastructure that will allow long-term human habitats on the Moon, to not only explore and study more of the Moon but also prepare for human exploration of Mars, In-Situ Resource Utilization (ISRU) is essential. The further humans go into deep space, the more important it will be to generate products with local materials. Robotics and transportation will be essential elements in this endeavor, as well as in-space manufacturing (ISM) techniques. We cover these aspects in the following sections.

### 4.4.1 Robotics and Transportation

The main goals of robotic exploration are to identify the best locations for future mining and habitat facilities, and to analyze the regolith content. Lunar robotics is and will be used to demonstrate technologies for human missions, perform surveys or sample returns for science, resource, and environment assessment, perform dangerous or risky tasks autonomously or with human remote control, and undertake construction of habitats and infrastructure.

The current stage of the exploration of the Moon is based on the use of semi-autonomous rovers and robotics for scientific research and location exploration. Remote-sensing technology is already utilized to investigate which deposits are where, and in what quantities. The lunar poles, in particular the south pole, are a practical place to set up bases and look for mining sites, for reasons of temperature stability and dependable solar energy, on top of the potential source of water ice in the permanently shadowed area of the craters, as shown in Figure 10. Moreover, as will be further detailed in the Habitation section, the use of lava tubes would potentially allow the development of safe habitats.

The balance of autonomous (based on AI or machine learning) and remote human control is another major factor affecting robotics design, with huge impacts on technology development, operational costs, capabilities, and speed of evolution toward a human outpost (with synergistic task sharing

between crew and robots). In fact, human intervention during this decade will be limited to sporadic, short-term presence on the Moon, to manage and interact with the existing robotic capabilities, and a heavy reliance on Earth remote control and post processing of data.

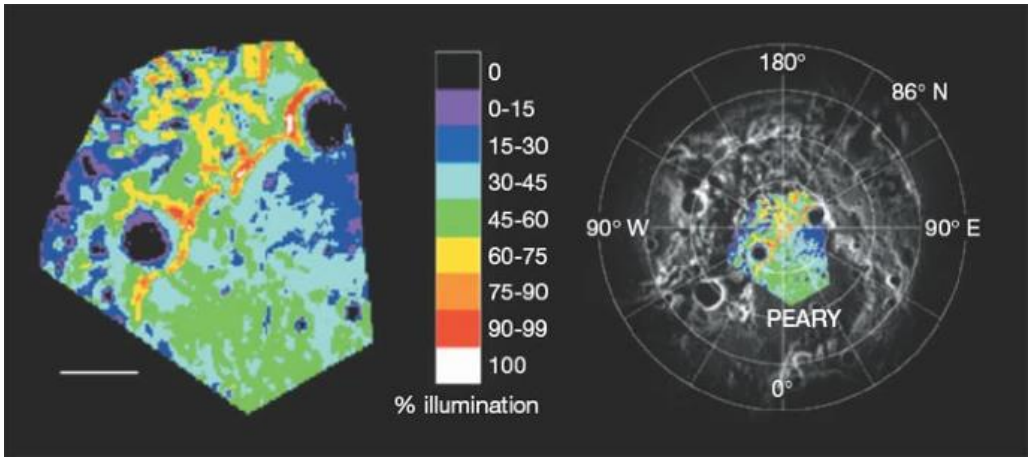



*Figure 10: Percentage of time a region of the lunar south pole is illuminated (Bussey, 2005)*

A mid-term evolution of such robotic capabilities might include:

- demonstrator missions, scaled up prototyping and testing of robotic machines used for autonomous exploitation of mining sites (drilling, material extraction and separation, manipulation, waste management);
- transportation of mined minerals and volatiles to the construction and habitation sites;
- construction of habitats, landing pads, power generation plants, transportation infrastructure, and shielding (see also In-Space Manufacturing section).

With the advancement in knowledge of locations and mining transport technologies, the need for more flexible robots, such as generic rovers, will decrease, allowing for more specialized and in-situ single-purpose robots. Increased power needs for exploration rovers, mining and transportation robots will also be supported by advances in power technologies (see Power section above).

The development of infrastructure to support robotics not ruggedized for the extreme environment will be helped by a first generation of robots used in the construction of these first structures. This will reduce the costs for future generations of robotic missions, especially benefiting the commercial and private sector initiative (Novara et al., 1998). Human presence during this phase will also be limited to the maintenance and supervision of robotic activity. Advanced science will predominantly be performed by humans in prebuilt labs.

Once the prototyping and testing phase is completed, the robotic technology will have to be scaled up to meet the demands for nominal production and a growing human presence in the lunar outpost. Hundreds of people would need water and oxygen, material for building new habitats, widespread transportation networks, food, and waste management. In this scenario, a fully interconnected system of autonomous robots will be used daily for supplying the settlement, transporting people and goods, building and repairing habitats and infrastructure, attending to farming facilities, and other tasks. The lunar settlement would become more self-sustainable and strongly dependent on automation. Human and machine interaction and task sharing would reach a balance point.

- **Challenges of Robotic Mining**

Mining on Earth is increasingly automated, done by robots remotely controlled by humans. The current technology in robotic mining on Earth is a useful basis for its application on the Moon, but a different environment presents huge challenges, as listed in Table 1 presented below. The same applies to the use of robots for the building of structures or the extraction of consumables such as water, oxygen, and nitrogen.

*Table 1: Earth-Moon environments - Comparison and Challenges (Adaptation of Mane (2022))*

| Environmental Factor | Earth | Moon | Impacts/Challenges of Lunar Conditions |
|---|---|---|---|
| ***Gravity at equator*** | 9.81 m/sec$^2$ | 1.62 m/sec$^2$ | Low gravity drastically changes structural requirements for buildings. |
| ***Rotation Time*** | 24 hours | 27.3 days | Latitude on the Moon determines effectiveness of solar power. Long night periods require high-capacity power storage. |
| ***Escape Velocity*** | 11.9 km/s | 2.4 km/s | Lower gravity makes it easier for spacecraft to launch. However, ejecta such as lunar dust can reach orbit more easily. |
| ***Mean surface temperature*** | 22°C | -153°C Night<br>107°C Day | Dealing with large temperature changes means materials will need to have low thermal expansion, and active thermal systems will need to draw from power supply to keep habitat modules at a comfortable temperature. |
| ***Temperature Extremes*** | -89°C to 58°C | -233°C to 123°C | |
| ***Heat Flow (average)*** | 63 mW/m$^2$ | 29 mW/m$^2$ | |
| ***Magnetic Field*** | Yes<br>(24-56 A/m) | No | Lack of magnetic field means space radiation can damage organic molecules and electronics. Various types of radiation protection are needed. |
| ***UV radiation*** | >300nm | >10nm (unfiltered spectrum) | |
| ***Ionizing Radiation*** | Minimal | Yes (SPE, GRC) | |
| ***Atmosphere*** | Yes | No | Inhabited modules must be protected from impacts that could breach their hulls and cause them to depressurize. External systems will need good protection or multiple redundancies. |
| ***Space debris/Meteoroids*** | No | Yes | |
| ***High Vacuum*** | No | Yes | Material must be strong enough to withstand atmospheric pressure from inside inhabited modules. Thermal systems cannot dissipate heat to the environment by conduction or convection. |

On the Moon, the strongly reduced gravity represents a huge advantage on different sides. For example, the support structures can bear higher loads than on Earth since the weight of the materials, a function of gravity, is lowered. Moreover, because there is partial gravity, ores and minerals of various densities will sink or rise as they do on Earth, so no special machines are needed to compensate for unusual microgravity physics (Lappa, 2004, Snell and Helliwell, 2005).

On the other hand, the absence of atmosphere can be a problem during drilling as this can cause excessive heat from friction that is not dissipated by the air flow. Similarly, blasting is extremely dangerous because projectiles would travel faster and farther then they would on Earth.

- **Future Recommendations**:

One of the first steps in any lunar robotic mission will be locating useful resources and determining their quantity and quality. To achieve this, it will be necessary to simultaneously explore multiple sites of high interest, based on existing orbital data indicating the presence of water ice and other resources. It is recommended that space agencies develop a combined strategy to coordinate exploration of various locations of interest concurrently and use this data to select the most promising sites for further characterization. Agency missions will increasingly benefit from private commercial support in developing and delivering novel instruments, so it is also recommended to enhance the public sharing of scientific data and to reinforce the economic framework to support the private initiative, by creating public-private partnerships to develop a space-based industry of in-situ resource utilization (Laurini et al., 2018).

The success of a lunar settlement will heavily depend on the efficiency and extension of the transport infrastructure and network. Transportation will start with simple rovers on wheels, but gradually evolve into fixed transportation systems like roads, railways or suspended cable car systems (Eugene and Wykes, 2010, Benaroya and Bernold, 2008). The latter will have the advantage, with respect to the Earth environment, of being able to carry people and loads over long distances on rough terrain very quickly, thanks to the absence of the air and thus reduced friction, and will also prevent the creation of dust clouds.

Short-term human and robotic exploration and resource exploitation will be limited to a radius of few dozen kilometers (Wanjek, 2020) around the base, mainly due to power limitations, but will eventually expand to bigger areas served by a growing network of connecting infrastructure, also relying on an increased power production (see In-Space Manufacturing and Power). This will also benefit a growing tourism industry (automated transportation and bigger power capacity will allow coverage of larger exploration zones) and the development of an economy of decentralized services such as power stations, launching pads, mining zones, heritage areas and science facilities (Wanjek, 2020).

### 4.4.2 In-Space Manufacturing

Long-duration exploration missions require a paradigm shift in the design and manufacturing of space architecture. The ability to perform In-Space Manufacturing (ISM) provides a solution towards sustainable, flexible missions (both in-transit and on-surface) through on-demand fabrication, repair, and recycling capabilities for critical systems, habitats, and mission logistics and maintenance (Litkenhous, 2019). For decades, LEO spaceflight relied heavily on orbital replacement units for system-based repair and maintenance. This model results in high mass maintenance requirements

and low sparing efficiency due to the lack of on-demand fabrication, repair, and recycling of consumables. To enable an Earth-independent logistics model this must be changed for long-duration cislunar missions and Moon settlement. In-space manufacturing and ISRU are thus critical, involving both manufacturing and recycling as a closed loop. ISRU technology includes the extraction of metal elements, water, or other resources from the surface soil and atmosphere of the Moon. A crucial component of ISRU is additive manufacturing (AM), which is the process of building something one layer at a time. It is expected that AM will be used to produce components for spacecraft and buildings using only materials found on the lunar surface. Investigation is also underway into the possibility of manufacturing solar cells using ISRU (see Power section) (Blue Origin, 2023a). The applications of AM in lunar construction are discussed further in the Habitation section of this report. A typical ISRU production plant consists of several subsystems: excavation, beneficiation (mineral processing to improve economic value), handling, oxygen extraction and purification, metal processing, gas liquefaction and storage, thermal control, and power.

The main challenge for human space exploration is resource availability. There are abundant resources on the Moon such as water ice, other volatiles, rare Earth elements, platinum group metals, and lunar regolith, which are the main sources of interest for ISRU (Crawford, 2015). The lunar regolith contains various minerals with major elements of oxygen, silicon, aluminum, calcium, iron, magnesium, and titanium.

Challenges on the Moon that must be addressed to enable ISM include microgravity, high vacuum, a complete lack of atmosphere, extreme radiation environment, seismic activity, and micrometeoroid impacts (Wang et al., 2022). More details on the differences between Earth and the Moon are illustrated in Table 1. For example, the temperature on the Moon varies between 100.15°K (–173°C) and 400.15°K (127°C), so the manufacturing facility must be able to remain rigid under these temperature ranges (Johnston et al., 2014). Due to the lack of atmosphere on the Moon, radiation shielding must be considered for all structural materials (Lindsey, 2004). For structural materials, especially metallic ones, the most promising technologies include additive manufacturing and the subsequent subtractive manufacturing, or machining and finishing.

Most structural materials used for spacecraft and manufacturing facilities are metals and alloys, among which steel, titanium, and nickel- and cobalt- based alloys are essential. Hundreds of tons of metals will be required for the initial exploration stages of the lunar surface (Guerrero-Gonzalez and Zabel, 2023). AM technologies for metallic materials are relatively well established under Earth's gravity and atmosphere. For example, jet engines, combustion chambers, and thrusters for launch vehicles have been successfully produced using additive manufacturing. To pave the way for the habitation of space, improving and adapting these technologies for ISRU is an important factor in reducing costs and logistical effort. ISM is to be done in a vacuum and microgravity, so all relevant technologies will need to be assessed under these conditions (Reitz et al., 2021). The high vacuum conditions on the Moon enable the additive manufacturing of reactive materials like titanium and magnesium without the strict atmosphere control necessary on Earth (Yang et al., 2022). Titanium is one of the metals that is used widely for aerospace and space sectors due to its balanced mechanical properties and high corrosion resistance. Magnesium shows the highest specific strength (strength to weight ratio) of almost all structural metals and has applications in aerospace, automotive and electronics industries (Lütjering et al., 2003, Sankaran and Mishra, 2017). These materials can be used for structural component printing and manufacturing with multiple purposes.

As mentioned previously, lunar regolith contains oxygen in addition to other desirable minerals. As part of oxygen extraction processes, the metals will become byproducts that could be leveraged for in-space manufacturing. (Guerrero-Gonzalez and Zabel, 2023, Licheri et al., 2022). Other ISRU processes include hydrogen reduction of ilmenite, carbonylation to produce low-carbon steels, molten regolith electrolysis to produce ferrosilicon alloys, and molten salt electrolysis – in particular the Fray-Farthing-Chen (FFC) Cambridge process – together with vacuum distillation, to produce aluminum–silicon alloys (Guerrero-Gonzalez and Zabel, 2023).

The AM of standard structural metallic materials like steel, titanium, and magnesium is relatively mature for on Earth manufacturing, with fine powders and thin wires as main feedstock. In the near-term phase of lunar settlement, feedstock materials and manufacturing facilities will need to be brought from Earth to the Moon to test the feasibility of using them there, and to optimize the processing sequence and material performance when manufactured under partial gravity and vacuum conditions. Experiments with AM machines aboard the ISS have confirmed that products manufactured in microgravity retain the properties of the substances they were manufactured from.

In the mid-term, if existing manufacturing facilities and materials can be supplied from Earth and used successfully, then the use of regolith and other lunar resources in AM can begin testing. This will require ISRU technology to be able to extract metals from the regolith to use as feedstock. In the long-term, it is recommended that lunar AM is utilized to build more manufacturing facilities, enabling the growth of a sustainable manufacturing program. This will allow any and all metal products to be created on the Moon without any import from Earth.

- **Lunar Habitats**

Lunar regolith can be exploited through additive manufacturing technologies and sintering to form construction elements such as bricks and shielding panels. The process involves mining and separating regolith, transporting materials to manufacturing sites, sintering, additive manufacturing, and reusing the materials to produce various products. Current AM technologies using lunar regolith as feedstock include extrusion-based printing, powder-based ink jetting, and selective laser melting (Wang et al., 2022). Each of these technologies have their own technical challenges, including organic compound binders being unstable under space radiation, the influence of heat transfer in vacuum sintering being unclear, and insufficient power supply. However, the shared major challenge is the final product being porous and brittle, making it unsuitable in extreme weather conditions and thermal cycles. Given the significant potential and importance, and the limited amount of lunar regolith collected on the Moon, to meet the pressing demand for scientific research and engineering applications, simulated and equivalent lunar soil based on minerals from Earth were formulated (Wang et al., 2022). However, all research findings obtained through this method need to be validated in the future using lunar regolith on the Moon. In the near-term, using simulated lunar regolith and manufacturing facilities from Earth on the Moon is achievable. Once the processes are better understood and validated, then in the mid-term, lunar regolith can be used together with manufacturing facilities brought from Earth for ISRU on the Moon. In the long-term, the goal is to process regolith in-situ to produce materials for in-space manufacturing of construction needs.

From a sustainability point of view, the additive manufacturing of lunar regolith has both positive and negative effects. For example, sintering of magnesium oxychloride (MOC, simulated lunar regolith from Earth minerals) cements for construction materials releases Hydrogen Chloride (HCl) and generates water (Osio-Norgaard et al., 2021). While water is a desirable by-product, systematic

collection and recycling of HCl as either input or media for certain processes needs to be well planned and executed.

- **Economic and Environmental Implications**

At some point, metals and other industrial resources will become too expensive to extract on Earth as a result of scarcity and increasing costs, and we could reach a point at which lunar materials are cheaper to extract. As for the environment, there's little debate that mining for resources causes irreparable damage to the Earth in terms of contamination, erosion, and other ecological impacts. A similar lunar concern might be mitigated by the absence of a lunar ecosystem or atmosphere. Promoting the use of the Moon (and asteroids) as mining locations for Earth needs in addition to lunar needs for self-sustainability and in support of a closed system motivates development of this technology for companies and economic development (Wanjek, 2020).

## 4.5 Habitation

- **Proposed Habitats - Background**

The design and construction of habitats depend on the mission length, objectives and the maturity of a program. This report defines a habitat as any controlled physical environment capable of sustaining human life which is surrounded by inhospitable conditions. Therefore, habitats for astronauts take many different forms such as space stations, spacecraft, capsules, pressurized rovers, and habitation modules.

The NASA Artemis missions are currently the most developed proposed missions to the Moon and therefore, they provide sufficient framework regarding the technological development of habitats for long-term missions to the Moon. Other agencies, companies and organizations have also proposed various missions to the Moon, and they will be included to provide a background of the near-term objectives and challenges to be addressed.

The USA, with assistance from other international space agencies and companies, plans to land on the Moon in 2025 with Artemis III, and then develop the Artemis Base Camp at the lunar South Pole (Kessler et al., 2022, NASA-Artemis3, 2023). In order to achieve this, the Orion spacecraft, which is a habitat that can hold up to 4 crew, has been designed for deep space operations and will dock with the Gateway space station (see Artemis). The Orion module is an example of temporary habitation that is sufficient for transporting astronauts to and from destinations in space, however it is not suited for long-term use and thus does not need to consider the challenges associated with long-term human survival in space, such as ISRU and renewable power (see ISRU and Power section).

The Gateway will be deployed around the Moon in a highly elliptical Near-Rectilinear Halo Orbit (NRHO) and is being developed, serviced and eventually utilized by NASA, ESA, JAXA, CSA, and various commercial partners (Mars, 2022). Initially, the Gateway will be autonomous and comprised of the Power and Propulsion Element (PPE), which will be supplied by Maxar Technologies, and the Habitation and Logistics Outpost (HALO) provided by Northrop Grumman and available from Artemis IV. ESA will provide the International Habitation (I-HAB) module which will be used in future Artemis missions. Astronauts will live and work in both HALO and I-HAB and together they allow stays of thirty days or more (Zaid, 2022). Therefore, Gateway will provide cyclic habitation capability that requires more long-term life support systems similar to what is used on the ISS, such as water recycling, however not the same extensive systems required on the surface of the Moon. Since the

Gateway is in orbit, Moon habitation challenges such as dust and moonquakes are not applicable, but temperature extremes and radiation must still be accounted for.

Once the Human Landing System (HLS) is added, the Gateway will serve as a stopover for human crews who will dock there on Orion, and use the HLS module to travel to and from the Moon's surface. SpaceX is contracted by NASA for their Starship HLS which will be the first lander to place Artemis astronauts on the Moon (Mahoney, 2020). Blue Origin and others are also competing to develop an alternate HLS, with the contract expected to be awarded in 2023 (Blue Origin, 2023b). The HLS is another example of a temporary human habitation module that does not require the rigorous systems needed for long-term human presence on the Moon. SpaceX's Starship, not to be confused with the Human Lander vehicle, is a large-scale human habitation spacecraft which is reusable and capable of transporting 100 people to the Moon and even Mars (Space X, 2020). Therefore, Starship will have to overcome many challenges of long-term space travel such as radiation and space weather events.

NASA's Artemis roadmap achieves permanent lunar habitation at the south lunar pole by the year 2031 (Amit Kshatriya, 2022). This habitat, the Artemis Base Camp (ABC), would consist of a single pressurized habitat module (Foundational Surface Habitat, FSH), an unpressurized lunar rover (Lunar Terrain Vehicle, LTV) for short-duration excursions, and a pressurized habitable rover (Habitable Mobility Platform, HMP) for long-duration excursions (NASA, 2020). The FSH and HMP are planned to support two crew each, and the FSH can support four personnel during crew rotation, or swap-outs, and can do so continuously for thirty days initially, potentially up to sixty days in the future (NASA, 2020). The ABC is envisioned as the first permanent foothold on the lunar surface, and support elements such as power generation and storage, communications, radiation shielding, a landing pad, and storage and waste disposal facilities, are slated to be added after the initial three components (Kessler et al., 2022).

### 4.5.1 Construction Challenges in the Lunar Environment

The near- and mid-term obstacles to future sustained human presence in the lunar environment include radiation exposure and space weather, lack of atmosphere, seismic activity (moonquakes), extreme temperature variations, micrometeoroids, and lunar dust (Benaroya, 2017). Each of these obstacles requires a specific mitigation strategy that must be integrated with any surface facility. There are many proposed methods for tackling these challenges, some of which are described below.

- **Radiation**

Space radiation and space weather are the biggest challenges to sustained human presence on the Moon. The Moon's lack of both an atmosphere and a magnetic field mean that radiation from various sources encounters no obstacles on its way to the lunar surface (Benaroya, 2017). Radiation will pose a twofold threat to any lunar habitat: firstly, it is capable of severing molecular bonds in organic compounds, causing severe health issues such as radiation poisoning and cancer (Kamiya et al., 2015); secondly, it can induce currents in conductors which may disrupt or damage electronic equipment such as solar panels, and may degrade the protective coatings of habitat modules (Benaroya, 2017). Additionally, the interaction of high-energy electrons with conductors may release x-rays which are given a head-start in bypassing any radiation shielding present (Benaroya, 2017). To

combat this, two different methods are proposed to protect astronauts against radiation and space weather events.

The first type of shielding exploits the lunar regolith itself. In-situ resource utilization can be used to collect lunar regolith which can be 3D-printed or sintered using additive manufacturing (AM) into shells around pre-built habitation modules by specialized robots (Altun et al., 2021). The thickness and density of these shells will determine the extent to which the shield blocks incoming radiation (Miller et al., 2009). This method has the advantage of using an abundant resource on the lunar surface, and may in future be achievable by a single robotic worker. The second method is to use electrostatic or electromagnetic force to redirect incoming radiation. Since all space radiation is to some extent affected by electric fields, using electrically charged plates as radiation shields has gained traction as a possible method of radiation shielding (Buhler, 2004). This method exhibits several drawbacks: it requires a continuous power supply to operate; it may interfere with electronics in and around the base; it may disturb the electrostatically charged lunar dust; and it requires all of its components to be brought from Earth and assembled (Buhler, 2004). For these reasons, lunar regolith, while not a perfect defense, is the more effective option of the two, and can be implemented as shielding in a variety of ways.

In the near future, polar surface habitats like the FSH may utilize the AM regolith shells discussed previously, and the technology to do this already exists (Altun et al., 2021). Alternatively, lunar habitats and buildings at the poles could burrow beneath the surface, utilizing the surrounding regolith instead of having to print a shell over the entire structure. Unfortunately, this option is hampered by the regolith itself – lunar soil becomes very dense below a depth of approximately 30 centimeters, and the partial gravity environment of the Moon would reduce the effectiveness of any current Earth excavation systems (Benaroya, 2017). Additionally, having the entirety of a habitat module underground would make maintenance almost impossible.

- **Temperature**

Thermal properties of the lunar habitat are also important. The temperature at the lunar surface oscillates between 100.15$^{o}$K (–173$^{o}$C) and 400.15$^{o}$K (127$^{o}$C), however this temperature range is less extreme near the lunar poles, due to their more consistent illumination (Benaroya, 2017). This wide range of temperatures is due to the absence of an insulating atmosphere (Benaroya, 2017). A habitat placed anywhere other than a PSR or a continually illuminated region at either of the poles will experience extreme thermal expansion and contraction over the course of a month, which is the time taken for the Moon to complete one rotation about its axis (Smith, 1993). This will induce stress in a habitat module which could rupture its hull and cause it to depressurize (Smith, 1993).

There are three main methods of mitigating this risk. The first is to construct the pressure vessel on Earth from materials which have low thermal expansion coefficients or are elastic enough to tolerate thermal expansion while holding shape and maintaining pressurization (Smith, 1993). This is already a major design principle in the development of any spacecraft (Brown, 2002). Secondly, the habitat can be landed in a place with a more stable temperature, such as a continuously illuminated region at the lunar south pole, which is the planned location of the Artemis Base Camp (NASA, 2020). The temperature is these regions is believed to hover around –50$^{o}$C (Bussey et al., 2005). The habitat can also be insulated with ISRU techniques. 3D-printing a regolith shell around a pre-built habitat would provide excellent thermal insulation, in addition to its radiation-blocking properties (Altun et al., 2021).

- **Seismic Activity and Meteoroids**

Moonquakes are a less severe issue than other hazards of the lunar environment. They tend to be weaker than earthquakes, and hence less able to damage structures, though shallow moonquakes are an exception (Seifamiri et al., 2022). However, some measure of protection is still necessary. Some proposed habitat designs include feet, which would improve the stability of the module against moonquakes (Kessler et al., 2022). The AM regolith shell discussed earlier would also provide additional stability, further insulating the habitat from the effects of moonquakes (Benaroya and Holt, 2018). However, the stresses on the regolith shell could cause it to crack in a manner that could not be completely fixed by the AM technology available.

The same techniques could be applied to protection against micrometeoroids. The Moon's lack of atmosphere means that meteoroids do not burn up before impacting the surface, making impacts by small objects a regular occurrence (Benaroya, 2017). Again, regolith shielding would go a long way towards mitigating the damage caused by an impact, but such impacts may be more likely than moonquakes to cause damage which cannot be completely fixed by the available AM systems.

- **Regolith and Lunar Dust**

Despite the positive aspects highlighted regarding the use of regolith, it also represents an obstacle to sustained human lunar presence. In its natural state, the regolith consists of fine, dust-like particles, which are shaped into miniscule, jagged, elongated grains by repeated meteorite strikes (Grün et al., 2011). The rough shape of these dust particles makes them sticky, and capable of damaging spacesuit material (Stubbs et al., 2007). All of the astronauts who walked on the Moon reported some difficulty with lunar dust, and its abrasive effects on their suits (Stubbs et al., 2007). Dust particles are also electrostatically charged by the photoelectric effect, which allows them to stick to surfaces by static cling (Benaroya and Holt, 2018). This gives the dust the ability to interfere with exposed electronics on the surface, and can cause it to get stuck in external systems (Stubbs et al., 2007).

Two methods of preventing dust abrasion are altering the outer wall of a habitat module, or preventing dust from electrostatically sticking to the skin of a such a module (Gaier et al., 2011, Jiang et al., 2020). The outer wall of a habitat, without the use of a regolith shell, can be designed as a layer of tiny cones, or similar shapes that reduce the contact area with any dust particles (Gaier et al., 2011). Alternatively, the skin of the habitat module can be made from a material with a similar work function (tendency to lose electrons) to the regolith, which prevents uneven charges from attracting dust to the surface (Jiang et al., 2020). This could once again be achieved by an AM regolith shell.

- **Recommendations for the Future**

Beyond these near-term challenges, the expansion of human presence on and around the Moon should progress as follows. It is this team's recommendation that infrastructure at the lunar south pole should continue to expand using modules sent from Earth. These modules, designed for human use, will contain equipment, robotics, and facilities such as laboratories, habitats, farming modules (see Life Support Systems section), and medical centers. Each structure will be shielded from radiation, micrometeoroids, and temperature changes by AM regolith shells. This should begin shortly after the completion of the ABC in 2031, and will continue until roughly 2040, when

infrastructure has developed sufficiently to allow the presence of a crew of eight to twelve individuals.

Following this, the expanded and permanent human presence on the Moon will allow for research into a greater degree of ISRU, and alternatives for permanent habitats. The prospect of building entire structures using AM with regolith is an attractive one, as it utilizes technology and infrastructure that will be present on the lunar surface by this time. However, the structural properties of this material are not well-understood. At the very least, building a test structure out of pure regolith on the Moon will determine if the structure is capable of remaining pressurized and stable against impacts and moonquakes. Even if the structure does hold, the airlock and similar interfaces will need to be imported from Earth.

As an alternative, lunar lava tubes are a possibility for sustained human presence. These structures were formed by the same volcanism that created the lunar maria, and are accessible from "skylight" holes in the lunar surface (Benaroya and Holt, 2018). Similar to the idea of buried habitats, the subsurface nature of the lava tubes offers many advantages. Firstly, the temperature in the lava tubes is believed to be much more stable than on the surface, around 253.15$^{o}$K (-20$^{o}$C), reducing the need for thermal insulation (Benaroya, 2017). Additionally, the thickness of the roofs of some larger tubes should provide ample protection against radiation and micrometeoroid impacts (Seifamiri et al., 2022, Benaroya and Holt, 2018). Being insulated from impacts, it is also believed that the interiors of lava tubes would be free from lunar dust, and the darkened interior prevents dust becoming charged by the photoelectric effect, though this would also prevent the use of solar power (Angelis et al., 2002). It has also been suggested that sections of lava tube could be sealed off and pressurized, creating a large, open environment in which humans could live with minimal additional protection (Martin and Benaroya, 2023).

This area is an intriguing opportunity for long-term research, which will be enabled by the ABC. However, it is not without its disadvantages. The lava tubes and skylights seem to be clustered around the boundaries between the mare and the lunar highlands, some distance away from either pole (Haruyama et al., 2009). This means that, aside from not being easily accessible from the ABC, the lava tubes may not have access to the frozen water at the lunar poles. However, by the time a habitat is successfully placed in a lava tube, it may be perfectly feasible to have a convoy of pressurized rovers transporting water from the ABC. Even so, life support systems will need to be more closed to deal with the lack of ready access to water.

### 4.5.2 Life Support Systems

Life support systems generally fall into one of three categories; open, closed, or hybrid. In an open system, all inputs and outputs are delivered to or removed from the site by external processes such as supply delivery, and manual waste removal. In a closed system, there are no external inputs or outputs and all resources and waste products are utilized and recycled in a continuous cycle. Hybrid systems are comprised of parts of both open and closed systems, and can vary significantly depending on the chosen solution for each subsystem – a spectrum ranging from fully open to completely closed. The primary challenges impacting life support systems on the lunar surface are supplying necessities and removing waste, and working towards reducing, and eventually removing, the reliance on resource deliveries from Earth. Lessons learned from creating a self-sustaining, permanent settlement on the lunar surface will also be critical to developing technology required to enable the longer-term goal of putting human boots on Mars.

With current life support technology, any temporary or longer-term lunar habitat will be wholly reliant on Earth for regular delivery of necessities like food, water, oxygen, fuel, and any other consumables needed for the duration of the mission. The longer the mission, the more difficult it would be to take all the necessary supplies in one vehicle, and more likely that resupply missions would be required to support the crew. More sustainable and regenerative life support systems are possible, and some components have already been developed for testing and use on the International Space Station (Miyajima, 2020). These systems are currently designed to support four to eleven crew for relatively short mission durations, and are backed up by regular deliveries from Earth. In order to achieve a sustainable permanent habitat, it is critical to improve existing life support systems, and develop cheaper, more self-sufficient closed systems, so that they can gradually replace the open, resupply-reliant approach that is currently utilized.

In the near term, it is logical to continue to leverage tried and tested life support systems that can be integrated into prefabricated habitat modules delivered from Earth. For example, the Sabatier system, whilst still needing development, uses a chemical reaction to convert carbon dioxide into methane and water, and is a reasonably well understood and reliable way to recycle one human waste product (NASA, 2011). Likewise, waste water reclamation processes are already used to reduce resupply mission launch weight, and onboard water carrying by 65% on the ISS (ESA, n.d.). By further developing this kind of hybrid life support system, it is possible to support a greater number of crew members for mid-length mission durations (3-6 months). However, iterative improvement and development of life support systems at the lunar settlement will be critical to ensuring a permanent human presence is possible. The peculiar challenges of the environment mean that even the technologies currently in use to support the crew of the space station may not be sufficiently robust. It also means that the only way to develop systems that will survive is to do so on the lunar surface, so they can be thoroughly tested there before they are implemented.

Once a permanent base of operations has been established, and there are regular crew rotations and reliable resupply schedules, the development of more self-sustainable life support systems can be addressed. Initially, the introduction of local oxygen generation from water – ideally obtained directly from the lunar surface – via electrolysis would reduce the reliance on oxygen deliveries from Earth. Additionally, the hydrogen byproducts of this process could be used in the manufacture of fuel cells (Miyajima, 2020). Electrolysis has been employed on the ISS since 2007 to produce oxygen from reclaimed water. This method eliminates many of the difficulties associated with delivery and storage of high pressure gaseous and liquid oxygen. Although a large-scale electrolysis generator would be more complex than simple gaseous oxygen storage tanks, and would require some technical maintenance, it is easier to achieve than the cryogenic storage requirements, and additional vaporization and pressurization needs of liquid oxygen (Chen and Sweterlitsch, 2022). Initial launch mass is considerably higher, but the extra cost is quickly offset by the elimination of the need for regular oxygen resupply (Miyajima, 2020).

The long-term goal is to create an environment that is as close as possible to self-sufficient; including the local production of food, fuel, and all other necessities for permanent lunar habitation by humans. One trade study (Miyajima, 2020) estimates that, for each crew member, approximately 27.3$m^2$ of crop growing space is required to sustainably produce food, biomass, cleanse water and carbon dioxide, and recycle human waste products, without any external supply input. The study assumes the local availability of water, oxygen, and nitrogen, but inputs could also initially be delivered from Earth and then replenished through the closed system waste management processes

as they are developed and improved. For a crew of six, it is estimated that approximately 9kW of power will be needed, which will need to be considered in the design of the settlement power systems (Miyajima, 2020).

### 4.5.3 Cooperation

Each step in expanding human presence beyond LEO relies on the readiness of new capabilities and technologies. As no single agency has the resources to develop all these critical capabilities, appropriately leveraging global investments in technology development and demonstration is important. Terrestrial commercial applications are well developed in several areas related to critical technologies. Fostering partnerships among government and commercial organizations will advance critical knowledge and accelerate the demonstration of spin-in technologies in space to validate the performance in a flight environment and identify unknown issues specific to operations beyond LEO. Partnerships can also lead to valuable experience and knowledge gap closures, especially in the development of standards and common interfaces (see Section 5 covering legal aspects). Coordination that supports development is an important tool for achievement of a global, strategic, coordinated, and comprehensive approach to space exploration.

### 4.5.4 Conclusion/Future Work

To summarize, the technological gaps that have been identified are presented in Table 2.

*Table 2: Set of technological gaps and recommendations*

| **Challenges and Technological Gaps** | **Recommendations** | | |
|---|---|---|---|
| | **Short Term** | **Mid Term** | **Long Term** |
| **Continuous Power Supply:**<br>- Environment: dust, radiation, thermal, day/night cycle.<br>- Cost to launch material, scalability. | Photovoltaic cells and regenerative fuel cells and batteries, and RTGs from Earth power up robots and rovers. | Scaling up the power generation by building more solar panels to support “on Moon” research.<br><br>Fission power systems in used. | Space solar cells continuously generating power.<br><br>Power generated by ISRU for a closed-loop fully operational power system. |
| **Energy storage:**<br>- Environment: dust, radiation, thermal<br>- Scalability | Using Li-ion batteries for backup and storage. | Expanding Li-ion backup capability using hydrogen fuel cells. | Manufacturing batteries on Moon when ISRU products are available. |
| **Power management and distribution:**<br>- Environment: dust, radiation, thermal.<br>- Dynamic power input and loads. | Self-contained micro grids connected to solar panel arrays. | Creating micro grids connecting local power sources. | Autonomous modular smart grids. |

| Challenges and Technological Gaps | Recommendations | | |
|---|---|---|---|
| | **Short Term** | **Mid Term** | **Long Term** |
| **ISRU Exploration and Exploitation:**<br>- Robotics autonomy and technological maturity<br>- Exploitation of consumables and construction material | Increase Robotic exploration capabilities. | Agencies and governments agreement on common exploitation and utilization mission.<br><br>Support for shared facilities and infrastructures.<br><br>Shared scientific database. | Full sharing of technologies and scientific data.<br><br>Full public and private cooperation. |
| **Locations and Transportation:**<br>- Legal agreements<br>- Technological advancement<br>- Environmental challenges (thermal, Dust, radiation)<br>- Investments | Propose lunar zones and safety aspects (e.g. dust, water contamination).<br><br>Increase research funding for prototyping and lunar demonstrators. | Sharing of infrastructure (e.g. transportation, power, landing pads) and facilities (e.g. hospitals, schools, farms, gyms). | Open access to common areas and facilities.<br><br>Interconnected outposts.<br><br>Full sharing of resources and infrastructures. |
| **ISRU:**<br>- ISM technologies and extraction technologies are not mature | On-Moon validation of experimental processes using simulated lunar regolith on Earth. | Using manufacturing facilities brought from Earth and lunar regolith. | Using manufacturing facilities produced on Moon and lunar regolith. |
| **Habitation:**<br>- Protecting humans from the lunar environment for sustained periods<br>- Highly effective radiation shielding<br>- Effective and practical dust mitigation | Preliminary habitat (ABC).<br><br>Begin building ISRU capability for AM regolith shells. | Expand south pole infrastructure to allow for more crew and greater variety of operations.<br><br>AM regolith shells for rigid and inflatable habitats.<br><br>Increase transport capability. | Fully established polar base.<br><br>Convoy of pressurized rovers for delivery of resources to/from launch sites and other habitats.<br><br>Investigation into lava tubes as permanent habitats. |

| Challenges and Technological Gaps | Recommendations | | |
|---|---|---|---|
| | **Short Term** | **Mid Term** | **Long Term** |
| **Consumables and Life Support:**<br>- Developing an effective mostly or fully closed life support system<br>- No fully closed life support systems yet exist<br>- Farming and agriculture in lunar soil | Leverage existing technologies for oxygen generation and waste management.<br><br>Test and validate developing technologies in upcoming missions. | Gradually transition subsystems between open and closed.<br><br>Continued development of more efficient closed subsystems.<br><br>Validation of lunar farming solutions. | Implementation of near-complete closed life support systems.<br><br>Oxygen generation and waste management from local ISRU.<br><br>Complete self-sufficiency. |

To progress from the current near-term to the intended mid- and long-term, the team has identified a set of incremental activities to address the environmental challenges and technological gaps outlined above.

Currently, solving the challenges of a sustainable human presence on the Moon's surface is still limited to the research and robotic exploration sector, and as of 2023 only one crew-rated mission has flown to the Moon (Artemis I) in this century. Nonetheless, the evolution of new mining techniques, additive manufacturing, solar power efficiency and production, along with new legal frameworks and an economic resurgence in commercial space is giving us an optimistic view of what a long-term future of sustainable presence, and a thriving lunar economy might look like.

# 5 Legal Framework

## 5.1 Overview

With the initiation of the Artemis Program and other lunar exploration interests, we have entered a new era in human ambition on the Moon. However, the current international legal framework which governs activities in outer space was drawn up last century, and although it established effective foundational principles, it has not been regularly updated with rules pertaining to recent developments. We believe the existing framework is insufficiently primed to tackle the most pressing contemporary issues surrounding expected activity on the Moon. We are presenting a 'think piece'; an incentivized, living document that incorporates all stakeholders, in a transitional and updated modern treaty, encompassing governance and compliance. In recognition of this, we have devised a new two-tiered legal framework, comprised of a treaty and an oversight committee, that addresses gaps in the current system and aids in delivering a vision for sustainable human presence on the Moon. It is worth noting that we do not move in contradiction to enduring principles that underline the existing framework, such as those outlined in the articles of the *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies* (hereafter "Outer Space Treaty" or "OST"). Ultimately, this report is intended as a thought experiment; a suggestion of what the system *could* look like, rather than as a complete package ready to be shoehorned directly into the real world, superseding existing treaty systems and international decision-making bodies. We recognize that applying the ideas herein to the current international space law landscape would be an iterative, staged process. Our proposed framework is presented below, alongside an accompanying rationale and a series of general recommendations.

## 5.2 Introduction

As the Artemis Program moves forward, it has become imperative to revisit the international legal instruments which regulate outer space. The Artemis Accords (chiefly formulated by the USA), and the ILRS (proposed by China and Russia), represent a turning point in the era of space exploration (de Zwart, 2022). The Outer Space Treaty was negotiated in a multilateral fashion (Blount, 2011). However, in 2020, the USA released the Artemis Accords (Smith, 2021b), a US-centric set of accords which other nations were invited to join after the Accords were written (Smith, 2021b). Thus far, 23 countries and one territory have signed the Accords (Lea, 2023).

While the Accords specify and detail how to implement the OST, they are still a US interpretation of the space treaties (Smith, 2021b). The need for the accords was to regulate the activities of relevant actors on the Moon and Mars (Deplano, 2021). This was necessitated by the fact that the OST gives a set of principles rather than specific guidance over actions (Smith, 2021b). The idea is that they will start a conversation with other partners and lead to an acceptable evolution of space law in due time (Deplano, 2021). However, the Accords have also received criticism. Firstly, the Accords purport to formulate international law (Smith, 2021b). Secondly, they are drafted not only for the Artemis program but also for the economic activities on the Moon, and the activities on Mars (Deplano, 2022). Thirdly, the Accords allow the creation of *de facto* "occupation" zones, known as 'safety zones', on the aforementioned celestial bodies, even though they do not allow sovereignty claims on the applicable celestial bodies (Gilbert, 2023). In light of the same, the Artemis Accords are not acceptable to all countries (Nelson, 2020).

The future of humanity includes activities on the Moon. It is imperative to formulate a new mechanism to regulate lunar activities. Such an international legal document should ideally be created in a multilateral manner so that it is acceptable to all nation-states (Baxter, 1965). Our aim is to create a codified dream which encompasses the vision of sustainable human presence on the Moon, and responsible utilization of lunar resources. In light of this, there is a necessity of an updated international treaty for outer space that addresses the legal complexities of furthering human exploration and other activities in the lunar domain. We acknowledge previous attempts at creating treaties, such as the OST, and examine legal developments and policy advances that need to be undertaken to ensure an effective treaty for the future. Such a treaty should be a living document that can address new issues as and when they arise.

We also recommend an engagement process for drafting a comprehensive treaty that properly reflects the diversity of stakeholders, and the complexity and dynamism of outer space operations and activities. We argue that the participation and engagement of all relevant actors, including states, commercial organizations, non-government organizations, and civil entities, is crucial for a working lunar governance system. The dynamic and evolving nature of lunar exploration, and the myriad of scenarios we could be faced with, are incorporated at the core of this framework with a mechanism that allows for iterative updates to the treaty.

Finally, we give general recommendations for further work, particularly towards points we were unable to resolve, including the matter of iteratively transitioning from the current system. The effective application of international law in this new and exotic domain is a colossal question, and we hope that our thought experiment below will advance this collective effort.

### 5.2.1 The current landscape: International Space Law

Space Law is a body of international treaties, agreements, conventions, United Nations General Assembly (UNGA) Resolutions, rules and regulations given by other international organizations, and, lastly, national rules that address the issue of utilization of outer space by human beings. Space Law can be described as the “body of law governing space-related activities” (Bozzo, 2018). The genesis of International Space Law was the launch of a Soviet satellite named Sputnik on 4 October 1957 (Cracknell and Varotsos, 2007). This was followed by the launch of the Explorer-1 satellite by the USA (Cracknell and Varotsos, 2007).

A ‘space race’ was to follow, and it became clear that there was no regulation in the area as this was beyond the scope of Air Law (Cheng, 1986). On 18 December 1958, the need for international cooperation and conventions for the peaceful use of outer space was recognized by the United Nations General Assembly (UNGA) (Sharma, 2022). The UNGA was “Wishing to avoid the extension of present national rivalries into this new field” (Sharma, 2022). It also established an *Ad Hoc* committee to deal with problems of law involved in space activities (Nelson, 2020).

Subsequently, a permanent body called the Committee on the Peaceful Uses of Outer Space (UNCOPUOS) was established on 12 December 1959 (Sharma, 2022). UNGA resolutions were adopted, which emphasized the peaceful use of outer space. Ultimately, the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, 1967 (Outer Space Treaty or OST) was written (Affairs, 2017). It formed the cornerstone of International Space Law and the four treaties which followed (Diederiks-Verschoor and Kopal, 2008).

A variety of matters are covered by International Space Law. These include the preservation of the space environment, settlement of disputes arising due to acts in space or of space objects, the rescue of astronauts, and international cooperation. Further, it also covers any liability owed by one nation to another for damage caused by space objects, as well as sharing of information about any potential threats in space (Freeland, 2010). The law of outer space purports to create a legal environment that would enable the achievement of the common goals of humankind (Sharma, 2022). These common goals and interests also include the peaceful utilization of outer space. Pertinently, space can be said to represent the applicability of the concept "common heritage of mankind" (Crawford, 2007)

### 5.2.2 Issues with contemporary international space law

The current international space treaties are decades old but still relevant. However, they do not directly cover new space activities (Lee, 2000) (Masson-Zwaan, 2023). The Moon Agreement addressed several points of contention, but space-faring nations have been reluctant to become parties to it (Vedda, 2017). This hesitation started from the moment the treaty was opened for ratification because the developing and the developed countries differed in their understanding of the term "common heritage of mankind" (Tronchetti, 2010). Although, the United Nations Office for Outer Space Affairs (UNOOSA) and COPUOS have been working on new international documents, they have only been able to come up with non-binding instruments. For instance, Masson-Zwaan has recommended an expansion of the lunar governance domain as the current treaties have gaps, and, at the same time, the new instruments are mostly 'soft laws' (Masson-Zwaan, 2023). Further, UN Space Treaties do not specifically address the contemporary activities expected to take place in cislunar space and on the Moon.

Recent attempts at creating a new treaty for outer space have received a tepid response (Hoffmann, 2019). Nevertheless, ensuring continued international collaboration in space through some effective system of international law remains a high priority. This fact is highlighted in a recent report from the Office of Inspector General of NASA (NASA, 2023a), which draws attention to the massive technical and fiscal burden (estimated USD $93 billion between 2012 and 2025 and hundreds of billions in the next two decades) that the agency must levy in support of the Artemis program, and the role international partners will play to ensure a viable, sustained presence on the Moon.

In lieu of a new treaty, and recognizing the benefits of cooperation and interoperability in lunar activities, in 2020 the USA drew up the Artemis Accords (Roulette, 2020). Grounded in a vision for the multilateral conduct of space exploration, the Accords essentially amount to a series of best-practice principles and mutual assurances and so, they are non-binding in nature (Deplano, 2021). Further, the Accords were formulated independently from the UN's multilateral treaty drafting mechanism, and, while grounded in the OST of 1967, the nature of the Accords has demanded innovative interpretations (Deplano, 2022).

No existing binding space law framework has sufficiently captured the full sectorial diversity of interested parties, which will be the primary stakeholders in the new lunar environment. There is a need to provide a general framework that is broad enough to encompass everyone's needs and that is open for future topics that cannot be predicted.

## 5.3 Our Core Recommendation: A Two-Tiered System

Any system that we devise to tackle the issues raised above should first be grounded in a low-resolution vision for the future state of lunar activities. In furtherance, we have grounded our work in the vision of what is of utmost relevance for our purposes. The legal framework and associated recommendations suggested by us are based in part on the international space law treaties which are in force. We have followed this approach to keep the practical aspects and principles of the treaties already in place, especially because they have found acceptance from most nations. We have also prioritized the primacy of states as they are the main benefactors of international law (Goldsmith and Posner, 2005). So, in our model, any treaty system (and agreements nested therein) can be signed and ratified only by recognized nation-states, accepting the axiom that the will of a state is communicated by its government which speaks on behalf of its people.

Further, the system proposed by us is built on two tiers. The tiers are built on a hybrid system based on a top-down and bottom-up approach. The upper tier will exclusively belong to states which will pass binding decisions. This tier will be based on an innovative treaty which draws its inspiration from the current working models of the Antarctic Treaty System and the Intergovernmental Agreement of the International Space Station (ISS) (Hanessian, 1960, 1998). In line with the aim of this report, we have laid emphasis on sustainability in our legal mandate. However, our legal provision of sustainability is of an enabling nature, and it is expected that once lunar activities become a reality, nations would reach an understanding as to the appropriate definition of this term.

The secondary tier shall be an international organization that will have an exclusive mandate over the Moon. It shall be composed of multiple stakeholders such as Nations, Companies, NGO's, Academia, and Employees Unions. This organization will have a broad mandate such as giving advisories and resolving disputes, while taking the views of all stakeholders into account. We would also like to concede that this organization is built on multiple working models such as the World Trade Organization (WTO), the International Labour Organization (ILO) and the United Nations (UN), although we have made some adjustments based on the criticism of those organizations.

### 5.3.1 Primary Tier: The Lunar Activities Treaty (LAT)

The idea of a working lunar treaty system presupposes that nation-states and other actors on the Moon will agree to work as a part of a multilateral system. This system also presumes that all the relevant entities are willing to engage in sustainable activities on the Moon. The latter presumption, especially, is based on relevant state practice as well as the public statements of relevant actors (Stenzel et al., 2018, Ellery, 2020).

For our purposes, and after discussion with our advisors, the Antarctic Treaty System (ATS) appears to be the most relevant, alongside the Inter-governmental Agreement of the ISS (IGA). There are multiple reasons for this. Firstly, the Antarctic Treaty System is a mature treaty which has existed since 1959 (Watts, 1992). It has a mechanism of revising its mandate and taking into account contemporary issues (Triggs, 2011). It also has much experience and has faced issues that are similar to what we could potentially face on the Moon (Lyall, 1998). For these reasons, the ATS presents a viable resource through which space law can be overhauled and lunar activities regulated.

On the other hand, the IGA is an excellent example of global cooperation against all odds (Kostenko, 2020). The nations which comprise the IGA and coordinate in the ISS are not necessarily the best of

friends, but they have been able to cooperate with each other, especially for scientific activities (Byers, 2019). This IGA system also allows governments to sign bilateral agreements permitting an expansion of the IGA beyond its mandate (Yakovenko, 1999). We are also impressed by the unique provisions dealing with Jurisdiction, 'Evolution', and Criminal Jurisdiction. We are mentioning these provisions in particular because they address unique points which were not done before their inception. Thus, our work will evolve the definitions and ideas that were propounded in these articles.

An adaptation of the ATS and IGA would allow States to give guidelines, agreements and new regulations regularly. This would also lead to the mitigation of new issues with their solutions being drawn up with new ideas and implementing them together in regular meetings that might happen, for example, annually. While the biggest contribution of the proposed system could be to mitigate the issues that already exist in space law, an even bigger impact would be the inclusion of new issues (as and when they arise) and the creation of a new multilateral treaty mechanism. This has worked well for Antarctica and while there would be reservations, it is a worthy model to consider especially in light of the current space law.

### 5.3.2 Primary tier: A working treaty system

The Primary Tier finds its genesis in the proposed Lunar Activities Treaty (LAT). The basic aspects of the LAT can be seen in Figure 11. Our draft treaty acknowledges that “humankind and the Celestial Bodies shall continue forever to be used exclusively for peaceful purposes and shall not become the scene or object of international discord.” It also declares that "Lunar Space shall be used for peaceful purposes only.” Further, the Treaty prohibits any measures of military nature, military maneuvers and testing of any type of weapons. Such a declaration has been deemed to be mandatory for us as there is an imminent danger of the militarization of outer space. We have also barred any dual use of civilian technology and use of weapons of mass destruction.




*Figure 11: Structure of the Lunar Activities Treaty*

It should also be emphasized that our proposed treaty categorically denies any claim or acknowledgement of sovereignty on the Moon. It shall also promote and bind the parties to conduct activities in a sustainable manner. The Treaty also acknowledges that there should be a mechanism for exchanging observers, scientific personnel, knowledge, and information between different parties to the Treaty. We believe that such a measure will lead to the expansion of knowledge. However, the provision will allow the restriction of access to locations that are deemed to be sensitive. Further, any such access shall be provided only on a voluntary and mutual basis.

A unique aspect of the Treaty is that it also defines a jurisdictional aspect of the Moon. This was deemed necessary as territorial jurisdiction is one of the key aspects of any such treaty. Another highly pertinent aspect of our proposed treaty are the annual meetings of all the parties to the treaty. The proposed system allows the state parties to make binding agreements and execute them among each other. This will keep the Treaty relevant and create a "living document" which would be contemporary for years to come. It is also conceded that this system is devised in the manner used under the Intergovernmental Agreement, as well as the Antarctic Treaty.

Readers may access the draft of the Treaty in Appendix B. The Treaty system proposed above shall be augmented by a new international organization which will form the second tier. The two tiers will work together to give binding decisions on parties. The methodology of how this would work is given in the next part.

### 5.3.3 Secondary Tier: Committee Overseeing Lunar Activities

Our proposed two-tier system encapsulates a secondary tier composed of an international organization (IO). This IO shall be responsible for the day-to-day functioning of activities on the Moon, and will have decision-making and dispute resolution mechanisms. However, it will remain secondary to the LAT and its decisions will become applicable only when they are ratified by the nations participating in the treaty. This rationale is important because nations are the primary actors in international law (Kumm, 2004).

Currently, there is no specific system in which all stakeholders in lunar activity can come together and address current and future topics. We believe there is a need to create a system in which the stakeholders can contribute to discussions on rule of law, code of conduct, and an agreed set of consequences. This demanded an unprecedented model for international co-operation, including all key stakeholders. The International Labour Organization (ILO) provided a context with which to explore such a model (Standing, 2010). After researching other similar bodies such as the UN Economic and Social Council (ECOSOC), and the International Astronautical Federation (IAF), we have deliberated and proposed the voting/membership model (Peterson, 2006, Mavroidis, 2022). Pertinently, ILO provides the ability for governments, employees, and employers to co-operate and have the capacity to act over their pursuits, instead of only a government as the sole agent in furtherance of its mandate. We were also inspired by the WTO dispute resolution mechanism in drafting this organization. The WTO is one of the very few organizations in the world with the capacity to make binding decisions on state parties. It has been a highly effective global trading mechanism since its genesis in 1995.

### 5.3.4 The suggested model of lunar governance organization

The Committee Overseeing Lunar Activities (COLA) shall be composed of three departments: the Secretariat, the Decision Making Council (DMC) and the Dispute Resolution Body (DRB). The brief description is given in Figure 12 below.

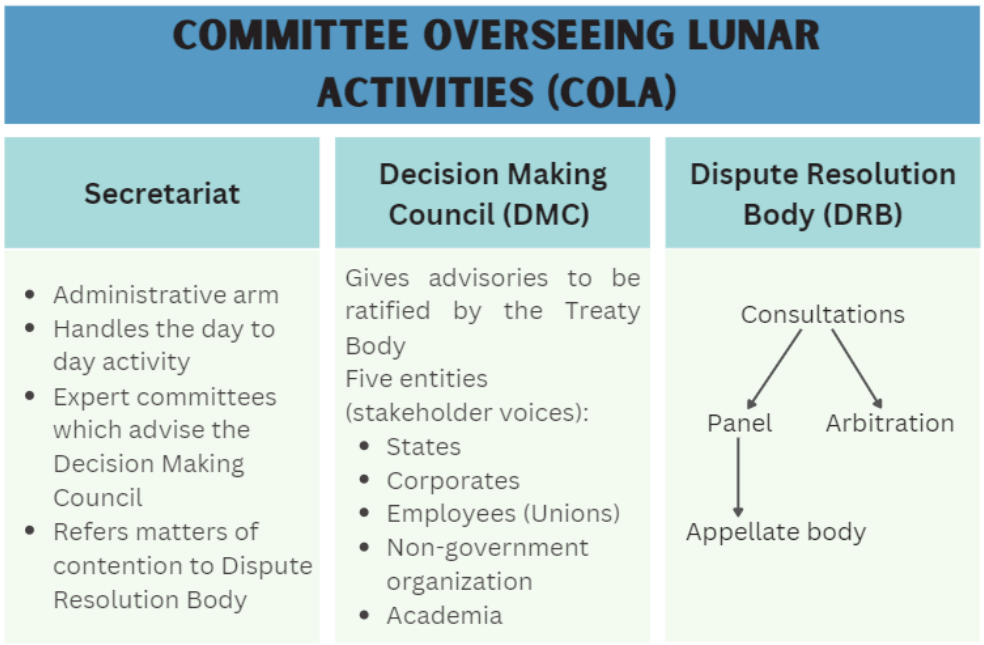



*Figure 12: Structure and functions of the COLA*

The first organ of the COLA is the Secretariat which manages the organization's daily activities. The need for this organ is justified by the reality that states can only participate in the DMC for a limited time each year. Therefore, the Secretariat will be composed of international civil servants. It will also suggest relevant topics of discussion to the DMC. Therefore, we deem that it will need experts from academia and space agencies. Such experts could work in committees, deliberate upon new topics, and give suggestions on topics that have been delegated to them by states. The dispute resolution mechanism shall be initiated and managed by the Secretariat.

The second organ is the DMC. It will act as the legislative system of the COLA. The DMC shall be responsible for passing advisories which will be forwarded to the LAT nations for ratification. It will also suggest topics to the Secretariat for further deliberation and expert opinion. Pertinently, the most unique aspect of the DMC is the presence of the five parties/groups. These include the following:

*Table 3: Groups in the DMC and justification for their inclusion*

| S.No. | Suggested party | Justification |
|---|---|---|
| 1 | Governments | States are the primary actors in international law. Current space programs are hugely supported by states. They will also be responsible for ratifying decisions of the DMC. So, their views need to be taken into account as they are major stakeholders. |
| 2 | Companies | Companies and private corporations are projected to drive the exploration and use of resources on the Moon, therefore it is necessary that they form part of the decision-making body on the Moon. |
| 3 | Non-Governmental Organizations | A non-governmental organization is an organization that generally is formed independent from government. The unique aspect of NGOs is their capability to take up issues which are of a broader nature and which would not be acceptable to economic benefactors. Their presence in the COLA will lead to highly holistic discussions on pertinent topics. |

| S.No. | Suggested party | Justification |
|---|---|---|
| 4 | Academics | Academic personnel, also known as faculty member or member of the faculty or academics or academic staff, shall also be a part of the DMC and provide expert opinions on the topics being deliberated. Their voice shall carry a lot of weight, and they will also be able to vote in the deliberations of the DMC. |
| 5 | Employees Trade Unions | Trade unions may be composed of individual workers and professionals exclusively working in companies and corporations. We believe that their presence is necessary in the DMC advisory voting system due to the current international human rights system, especially the International Bill of Human Rights. |

The groups and the rationale for adding them to the DMC is given in Table 3 above. We deem that including these five entities is necessary for a holistic lunar governance mechanism. Each group mentioned in the table above shall have one vote. The group will appoint one representative to give its views.

The final organ of COLA is the Dispute Resolution Body (DRB). The DRB enforces and deals with the idea of compliance and sanction. We believe that disputes on the Moon will arise and so, require a binding mechanism. As stated previously, our proposed mechanism closely follows the dispute resolution system at the WTO. However, we do concede that the consensus-based system at the WTO has currently led to a stalemate at the Appellate Body of the WTO (Kerremans, 2022). The basic system of the DRB has been given in Appendix C. Thus, we have strived to avoid these issues while incorporating the Dispute Settlement Understanding. We have also added an optional mechanism of arbitration similar to the WTO model.

### 5.3.5 An appraisal of the two-tiered system

The incentives for signing a two-tiered Treaty system of international co-operation, to further lunar activities in a hostile space environment include:

1. Protection of assets – however, claiming a section as a safe zone first does not result in ‘ownership’ or ‘jurisdiction’.
2. Access to equitable share of in-situ resources and facilities of others (for example, medical facilities, communications, lunar lander).
3. All stakeholders have the capacity to voice their opinion on parties of the Treaty and COLA via their nominated representative, have agency and voting rights over their objectives and goals in line with international co-operation, in designing a code of conduct. Whether it be an NGO, corporation, or trade union, the stakeholders have the ability to rule on their own goals and objectives and report annually to Treaty States, over concerns, update, and progress to the treaty. This is unique because while generally only States are liaised with in this process, our model is giving a level of voice and agency to all parties invested in lunar activity
4. All parties to lunar activity have a means of addressing grievance in terms of assets and capability interruption within an international accord.
5. All parties to lunar activity have the ability to make legal points as part of a council to enhance their goals and objectives, and revenue earning capacity.

6. Due to the nature of operating in a hostile environment, all parties of lunar activity have an added incentive to be inclusive of international co-operation, which can co-ordinate rescue efforts in faster timeframes, with a larger asset pool of spacecraft etc.
7. Pursuit of science with international collaboration to further sustainable, safe human exploration of lunar activities and space exploration.
8. Mutual reassurances of nonmilitary dual use, according to the principle of no "weaponization".
9. Opportunity to coordinate the preservation of landscape for scientific, cultural and aesthetic purposes.
10. Benefiting from standardized and agreed-upon procedures for safe lunar activity procedures.

## 5.4 Conclusion

This is not a comprehensive model that answers all questions. It is rather a place for all interested players to start a conversation towards a long-term human presence on the Moon, which they can review annually, with scope and change as is indicative of the passage of time. The code of conduct written by the COLA and membership, provides an opportunity for safe, sustainable, profitable lunar use today and for preserving it into the future. We believe that this model will be able to support the activities on the Moon until 2050 and create a holistic mechanism for the same.

# 6 General Conclusions

After exploring the history of lunar exploration and the current status of crewed missions to the Moon, the team identified the various factors influencing upcoming lunar missions, such as growing technology, economic feasibility, and international and legal issues. We also highlighted the potential geopolitical concerns that could arise as a result of the competitions between Artemis and ILRS and the need to address them to ensure a sustainable human presence on the Moon. To achieve this goal, we proposed an approach based on technological development and the creation of a new legal framework that can manage international and commercial conflicts.

On the technological side, the three aspects that were identified as having the greatest impact to the physical sustainability of a lunar settlement were power, in-situ resource utilization and human habitation. The team explored the challenges and technological gaps associated with these resources and provided recommendations that build a roadmap for achieving long-term presence of humans in cislunar space and on the Moon. With the advancement of solar power efficiency and production, additive manufacturing, habitation and life support systems, the long-term future of sustainable presence, and a thriving lunar economy can be achievable.

As for the legal part, the team has focused on the contemporary situation of international space law. After an extensive analysis of the treaties and contemporary literature, we found that the current space law is lacking in many respects. The most recent space agreement, the Artemis Accords is in the form of a guideline and also a unilateral initiative. Subsequently, the team looked at multiple international law models across the globe to find a working system of international space law. The Antarctic Treaty and the ISS Intergovernmental Agreement were chosen as the basis of a treaty system, while multiple entities such as the World Trade Organization, the United Nations and the International Labour Organization have formed the basis of the proposed international organization. The two-tier structure suggested by us keeps the initiative with nation-states, while giving voice to other stakeholders. Ultimately, the aim of the proposed mechanism is to provide a working system which can be of relevance in the future.

The road to sustainable human settlement on the Moon requires a concerted effort in terms of technology development and international cooperation, and this report provides a roadmap and recommendations for achieving this goal. By addressing the challenges of power, resource utilization, and human presence, and revising the international legal framework for cislunar activities, we pave the way for a peaceful and prosperous future on the lunar surface, marking a new era of exploration and discovery for humankind.

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# Appendices

## Appendix A

*Table 4: Lunar robotic missions of private companies*

| Mission Name | Private Organisation | Objectives | Goal Year | Launch Vehicle | Status | Origin Country | Type |
|---|---|---|---|---|---|---|---|
| HGS-1 | Hughes | Satellite | 1997 | Proton-K | Success | USA | Flyby |
| Hakuto R-1 | ispace | Lander | 2023 | Falcon 9 | In Progress | Japan | Primary Mission |
| Beresheet 1 | SpaceIL | Lander | 2019 | Falcon 9 | Failed | Israel | Sole Mission |
| Manfred Memorial Mission | LuxSpace | Orbiter | 2014 | Long March 3C | Success | Luxembourg | Payload |
| Mission One | Astrobotic Technology | Lander | 2023 | Vulcan Centaur | In Progress | USA | Conglomerate |
| Colmena | UNAM | Rover | 2023 | Vulcan Centaur | In Progress | Mexico | Conglomerate |
| IM-2 | Intuitive Machines | Lander | 2023 | Falcon 9 | In Progress | USA | Conglomerate |
| Australia Lunar Exploration Mission | Large Group | Rover | 2023 | TBD | In Progress | Australia | Conglomerate |
| IM-3 | Intuitive Machines | | 2024 | | | | |
| Blue Ghost | Firefly Aerospace | Lander | 2024 | Falcon 9 | In Progress | USA | Conglomerate |
| Hakuto R-2 | ispace | Lander, Rover | 2024 | Falcon 9 | In Progress | Japan | Sole Mission |
| Mission 3 | ispace | Lander | 2024 | TBD | In Progress | Japan | Sole Mission |
| Ardoride | Momentus Space | Orbiter | 2024 | TBD | In Progress | USA | Conglomerate |
| Blue Moon | Blue Origin | Lander | 2024 | TBD | In Progress | USA | Sole Mission |
| Beresheet 2 | SpaceIL | Lander | 2025 | TBD | In Progress | Israel | Sole Mission |
| Garatea-L | Airvantis | Orbiter | 2025 | PSLV-C11 | In Progress | Brazil | Conglomerate |

## Appendix B

**Draft Treaty on the Lunar Activities in Outer Space, 2023 [or the Lunar Activities Treaty (LAT)]**

Recognizing that it is in the interest of all humankind that Outer Space and the Celestial Bodies shall continue forever to be used exclusively for peaceful purposes and shall not become the scene or object of international discord;

Acknowledging the substantial contributions to scientific knowledge resulting from international cooperation in scientific investigation in Outer Space;

Convinced that the establishment of a firm foundation for the continuation and development of such cooperation on the basis of freedom of scientific investigation on the Moon is in the interests of science and the progress of all humankind;

Convinced also that a treaty ensuring the use of the Moon for peaceful purposes only and the continuance of international harmony in Outer Space will further the purposes and principles embodied in the Charter of the United Nations;

Have agreed as follows:

ARTICLE I

The Moon shall be used for peaceful purposes only. There shall be prohibited, *inter alia*, any measures of a military nature, such as the establishment of military bases and fortifications, the carrying out of military maneuvers, as well as the testing of any type of weapons.

ARTICLE II

Freedom of scientific investigation on the Moon and cooperation toward that end, as per the provisions of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, shall continue, subject to the provisions of the present Treaty.

ARTICLE III

1. The object of this Treaty is to establish a long-term international cooperative framework among the States, on the basis of genuine partnership, for the detailed design, development, operation; and utilization of a permanently inhabited civil lunar base for peaceful purposes; in accordance with international law. This lunar base will enhance the scientific, technological, and commercial use of outer space.

2. This Treaty specifically refers to the activities on the Moon, the nature of partnerships, including the respective rights and obligations of the Contracting States in this cooperation.

3. This Treaty further provides for mechanisms and arrangements designed to ensure that its object is fulfilled. In order to promote international cooperation in scientific investigation and other activities on the Moon, as provided for in Article II of the present Treaty, the Contracting Parties agree that, to the greatest extent feasible and practicable:

(a) information regarding plans for scientific programs in Outer Space shall be exchanged to permit maximum economy and efficiency of operations;

(b) no unilateral action shall be taken by any party which jeopardizes any provision of this treaty or the interest of mankind.

(c) full and complete cooperation with the Committee Overseeing Lunar Activities (COLA) will be ensured.

4. In implementing this Article, every encouragement shall be given to the establishment of cooperative working relations with those Specialized Agencies of the United Nations and other international organizations having a scientific or technical interest in Outer Space, including but not limited to, the United Nations Office of Outer Space Affairs (UNOOSA) and the Committee on Peaceful Uses of Outer Space (COPUOS).

ARTICLE IV

No acts or activities taking place while the present Treaty is in force shall constitute a basis for asserting, supporting or denying a claim to territorial sovereignty on the Moon or create any rights of sovereignty on the Moon.

ARTICLE V

1. Any nuclear explosions on Moon and the disposal there of radioactive waste material shall be prohibited.

2. Any dual use of civilian nuclear energy or any other sources of energy shall be prohibited.

3. All kinds of weapons shall be prohibited in cislunar space and the Moon.

4. In the event of the conclusion of international agreements concerning the use of nuclear energy, including nuclear explosions and the disposal of radioactive waste material, to which all of the Contracting Parties whose representatives are entitled to participate in the meetings provided for under Article IX are parties, the rules established under such agreements shall apply on the Moon.

ARTICLE VI

The Treaty shall apply to all activities on the Moon.

ARTICLE VII

1. In order to promote the objectives and ensure the observance of the provisions of the present Treaty, each Contracting Party whose representatives are entitled to participate in the meetings referred to in Article IX of the Treaty shall have the right to designate observers to carry out any inspection provided for by the present Article. Observers shall be nationals of the Contracting Parties which designate them. The names of observers shall be communicated to every other Contracting Party having the right to designate observers, and like notice shall be given of the termination of their appointment.

2. The Committee Overseeing Lunar Activities shall also have the right to designate observers to carry out the above function.

3. All areas of cislunar space and the Moon, including all stations, installations and equipment within those areas, and all spacecrafts at points of discharging or embarking cargoes or personnel, shall be open to inspection by any observers designated in accordance with paragraph 1 of this Article.

4. Each Contracting Party shall, at the time when the present Treaty enters into force for it, inform the other Contracting Parties, and thereafter shall give them notice in advance, of

(a) all expeditions to and within the Moon, on the part of its spaceships or nationals, and all expeditions to the Moon organized in or proceeding from its territory; and

(b) all stations on the Moon occupied by its nationals.

5. The Contracting Parties shall remain liable in accordance with the Liability Convention.

6. In the event of a claim arising out of the Liability Convention, the Contracting Parties shall consult promptly on any potential liability, on any apportionment of such liability, and on the defense of such claim.

ARTICLE VIII

1. In order to facilitate the exercise of their functions under the present Treaty, observers designated under paragraph 1 of Article VII and scientific personnel exchanged, and members of the staffs accompanying any such persons, shall be subject only to the jurisdiction of the Contracting Party of which they are nationals in respect of all acts or omissions occurring while they are on the Moon for the purpose of exercising their functions.

2. Parties concerned in any case of dispute with regard to the exercise of jurisdiction on the Moon shall immediately consult together with a view to reaching a mutually acceptable solution through the Committee Overseeing Lunar Activities.

ARTICLE IX

1. Representatives of the Contracting Parties named in the preamble to the present Treaty shall meet within two months after the date of entry into force of the Treaty, and thereafter annually, for the purpose of exchanging information, consulting together on matters of common interest pertaining to the Moon and cislunar space, and formulating and considering, and recommending to their Governments, measures in furtherance of the principles and objectives of the Treaty, including measures regarding:

(a) use of the Cislunar Space and the Moon for peaceful purposes only;
(b) facilitation of scientific research in Cislunar Space and the Moon;
(c) facilitation of international scientific cooperation in Cislunar Space and the Moon;
(d) facilitation of the exercise of the rights of inspection provided for in Article VII of the Treaty;
(e) questions relating to the exercise of jurisdiction in Cislunar Space and the Moon;
(f) preservation and conservation of living resources in Cislunar Space and the Moon;
(g) Non-military use of Cislunar Space and the Moon;
(h) sustainable use of Cislunar Space;
(i) any other issue that the member states of the treaty deem to be relevant.

2. Each Contracting Party which has become a party to the present Treaty by accession under Article XIII shall be entitled to appoint representatives to participate in the meetings referred to in paragraph 1 of the present Article, during such time as that Contracting Party demonstrates its interest in Outer Space by conducting substantial scientific research activity there, such as the establishment of a scientific station or the dispatch of a scientific expedition.

3. Reports from the observers referred to in Article VII of the present Treaty shall be transmitted to the representatives of the Contracting Parties participating in the meetings referred to in paragraph 1 of the present Article.

4. All advisories of the Decision Making Body of the Committee Overseeing Lunar Activities shall be tabled and transmitted to the Contracting Parties for ratification and enforcement.

5. Decisions of the Dispute Resolution Mechanism of the Committee Overseeing Lunar Activities shall be transmitted to the Contracting Parties for ratification and enforcement

6. The meetings shall take place exclusively on planet Earth.

7. The measures referred to in paragraph 1, 4 and 5 of this Article shall become effective when approved by all the Contracting Parties whose representatives were entitled to participate in the meetings held to consider those measures through a majority.

8. Any or all of the rights established in the present Treaty may be exercised as from the date of entry into force of the Treaty whether or not any measures facilitating the exercise of such rights have been proposed, considered or approved as provided in this Article.

ARTICLE X

1. The Committee Overseeing Lunar Activities (hereinafter referred to as “the COLA”) is hereby established.

2. The COLA shall provide the common institutional framework for the conduct of activities in Cislunar space and on the Moon.

3. The parties of LAT and COLA shall strive to work harmoniously with each other, with the outlying aim to benefit all of humanity.

ARTICLE XI

Each of the Contracting Parties undertakes to exert appropriate efforts, consistent with the Charter of the United Nations, to the end that no one engages in any activity in Outer Space contrary to the principles or purposes of the present Treaty.

ARTICLE XII

1. If any dispute arises between two or more of the Contracting Parties concerning the interpretation or application of the present Treaty, those Contracting Parties shall consult among themselves with a view to having the dispute resolved by negotiation, inquiry, mediation, conciliation, arbitration, judicial settlement or other peaceful means of their own choice.

2. If the dispute is not resolved within three months, then the Contracting Parties shall refer the dispute to the International Court of Justice (ICJ) for settlement. The decision of the ICJ will be final and binding.

ARTICLE XIII

1. (a) The present Treaty may be modified or amended at any time by unanimous agreement of the Contracting Parties whose representatives are entitled to participate in the meetings provided for under Article IX. Any such modification or amendment shall enter into force when the depositary Government has received notice from all such Contracting Parties that they have ratified it.

(b) Such modification or amendment shall thereafter enter into force as to any other Contracting Party when notice of ratification by it has been received by the depositary Government. Any such Contracting Party from which no notice of ratification is received within a period of two years from the date of entry into force of the modification or amendment in accordance with the provisions of subparagraph 1(a) of this Article shall be deemed to have withdrawn from the present Treaty on the date of the expiration of such period.

2. (a) Any of the Contracting Parties whose representatives are entitled to participate in the meetings provided for under Article IX so requests by a communication addressed to the depositary Government, a Conference of all the Contracting Parties shall be held as soon as practicable to review the operation of the Treaty in the year 2050.

(b) Any modification or amendment to the present Treaty which is approved at such a Conference by a majority of the Contracting Parties there represented, including a majority of those whose

representatives are entitled to participate in the meetings provided for under Article IX, shall be communicated by the depositary Government to all the Contracting Parties immediately after the termination of the Conference and shall enter into force in accordance with the provisions of paragraph 1 of the present Article

(c) If any such modification or amendment has not entered into force in accordance with the provisions of subparagraph 1(a) of this Article within a period of two years after the date of its communication to all the Contracting Parties, any Contracting Party may at any time after the expiration of that period give notice to the depositary Government of its withdrawal from the present Treaty; and such withdrawal shall take effect two years after the receipt of the notice by the depositary Government.

ARTICLE XIV

1. The present Treaty shall be subject to ratification by the signatory States. It shall be open for accession by any State which is a Member of the United Nations, or by any other State which may be invited to accede to the Treaty with the consent of all the Contracting Parties whose representatives are entitled to participate in the meetings provided for under Article IX of the Treaty.
2. Ratification of or accession to the present Treaty shall be effected by each State in accordance with its constitutional processes.
3. Instruments of ratification and instruments of accession shall be deposited with the Secretary-General of the United Nations, hereby designated as the depositary.
4. The depositary shall inform all signatory and acceding States of the date of each deposit of an instrument of ratification or accession, and the date of entry into force of the Treaty and of any modification or amendment thereto.
5. Upon the deposit of instruments of ratification by all the signatory States, the present Treaty shall enter into force for those States and for States which have deposited instruments of accession. Thereafter the Treaty shall enter into force for any acceding State upon the deposit of its instrument of accession.
6. The present Treaty shall be registered pursuant to Article 102 of the Charter of the United Nations.

ARTICLE XV

The present Treaty, done in the Afrikaans, English, French, German, Hindi, Italian, Russian Italian, each version being equally authentic, shall be deposited in the archives of the United Nations, which shall transmit duly certified copies thereof to the Governments of the signatory and acceding States.

## Appendix C

**Understanding on rules and procedures governing the settlement of disputes**

Members of the Committee Overseeing Lunar Activities hereby agree as follows:
The rules and procedures of this Understanding shall apply to disputes brought pursuant to any dispute or disagreement between the members of the Committee Overseeing Lunar Activities (hereafter COLA).

The Dispute Settlement Council is hereby established to administer these rules and procedures. The body shall be created and managed by the Secretariat under the rules which it shall prepare in consonance with the Decision-Making Council.

In case of any dispute or disagreement, the parties will initiate consultations with each other after informing the Secretariat. If no resolution occurs within thirty days, the parties can then choose formal proceedings or go for arbitration. In case of arbitration, secrecy can be maintained, and parties shall be bound by the decision of the arbitrators. If no such arbitration agreement is forthcoming within two weeks, the parties will then inform the Secretariat which will initiate formal dispute resolution proceedings. It should be emphasized that the parties can skip arbitration completely and start the formal proceedings at the COLA immediately.

The first stage shall be the formation of a panel, which will be composed of five expert members drawn from the Secretariat. These experts shall have the specialization required in the particular case. Once the panel is convened, it shall give binding decisions within three months. Once the decision is made, it shall be circulated as a report to the parties to the dispute, as well as the members.

If any party to the dispute is not satisfied with the decision, it will have the right to file an appeal with the Appellate Body (AB). The AB shall constitute five experts appointed by nations, and its decision shall be final and binding. The parties shall also have the right to appoint one judge each of their choosing in the AB. In case there is a need for enforcement of the decision, then Treaty members can be approached to ratify and enforce the decision.

**COLA Eligibility, membership and voting rights**

[The first 2 years temporary, must provide evidence of product, service, technology, and industry applied in space during active service, 2 years grace]

1. Parties hereby agree that any membership of companies/corporates shall be based upon the entity's project which it aims to conduct on the Moon.
2. Parties hereby understand that any research project or establishment of the same, is a time taking process, and therefore a temporary membership will be given to such an entity for parties participating in discussions related to its projects. Holders of a temporary membership shall have the status of an Observer only, and no voting rights.
3. Parties also agree that, in case the entity is unable to fulfill its mandate within a period of 2 years, its membership shall be revoked and it will be granted the Observer status. The Observer status will continue in perpetuity, provided that the entity continues to pay Observer's fees as directed by the secretariat from time to time.
4. If the entity is able to achieve its mission parameters, then it shall be given full membership of the organization.
5. Provided that if an entity is part of a member states' space project or has received a contract for a member state, then it shall automatically be granted the temporary membership.

We believe these criteria are necessary in order to suggest a starting point for guidelines of eligibility for all parties that are stakeholders in lunar activities. The eligibility of such parties is due to their significant investment, scientific research and infrastructure necessitating international co-operation in a hostile environment.

FINAL REPORT - SOUTHERN HEMISPHERE SPACE STUDIES PROGRAM 2023

# JOURNEY TO THE MOON

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